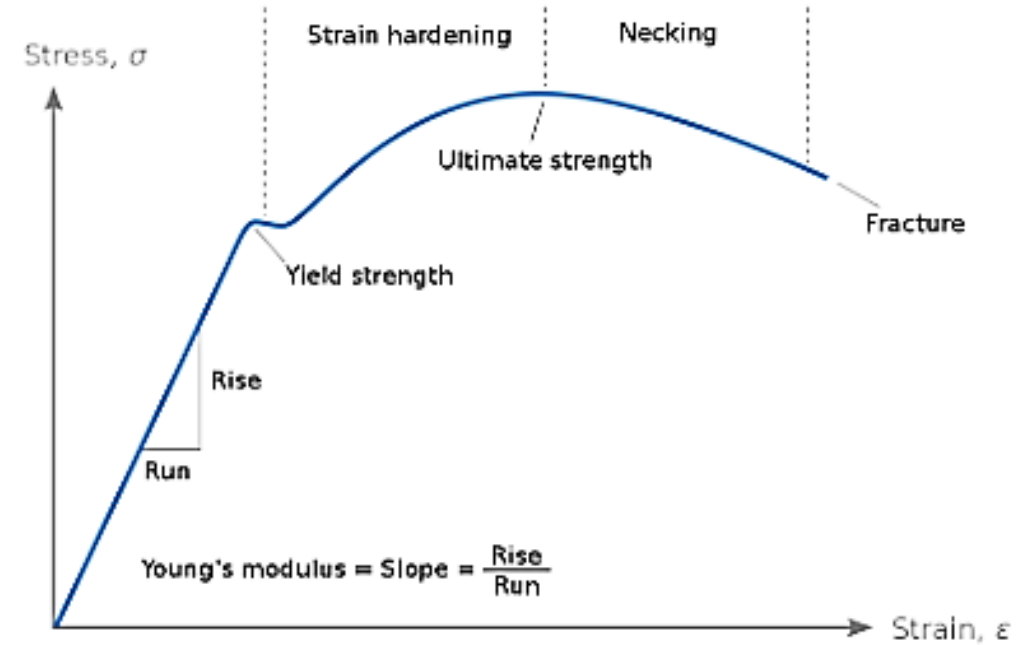
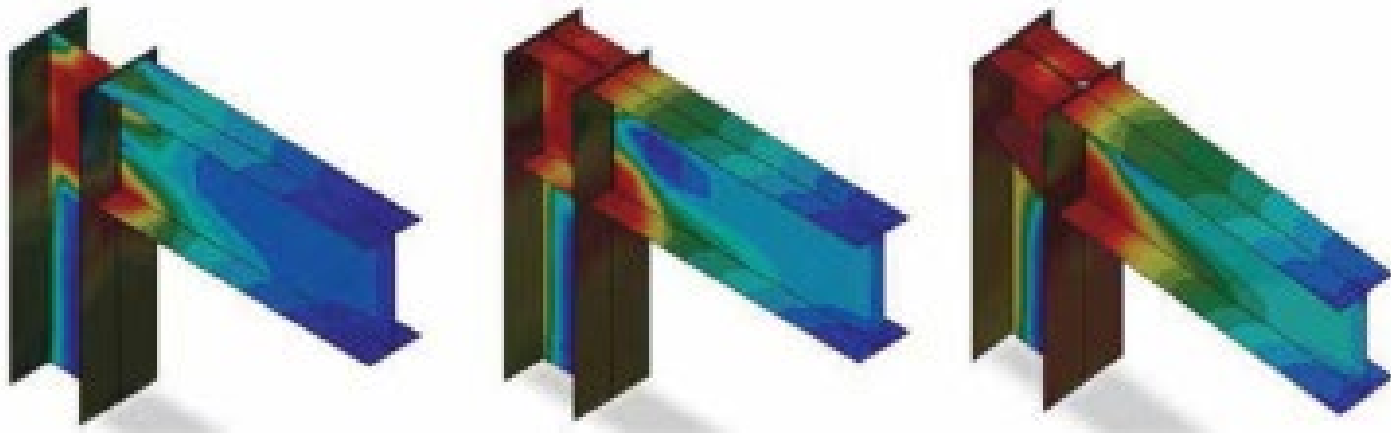
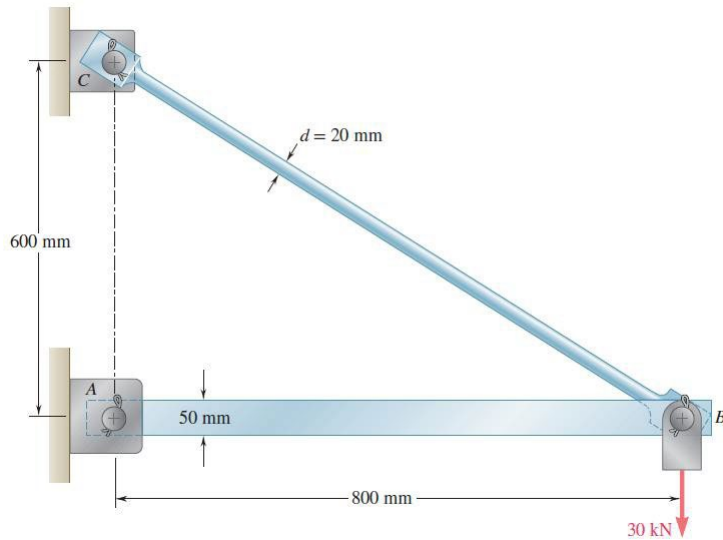


Chapter One: Stress and Strain



REVIEW OF THE METHODS OF STATICS



Consider the structure shown in Fig. 1.1, which was designed to support a 30-kN load. It consists of a boom AB with a 30×50 -mm rectangular cross section and a rod BC with a 20-mm-diameter circular cross section. These are connected by a pin at B and are supported by pins and brackets at A and C , respectively. First draw a *free-body diagram* of the structure. This is done by detaching the structure from its supports at A and C , and then showing the reactions that these supports exert on the structure (Fig. 1.2). Note that the sketch of the structure has been simplified by omitting all unnecessary details. Many of you may have recognized at this point that AB and BC are *two-force members*. For those of you who have not, we will pursue our analysis, ignoring that fact and assuming that the directions of the reactions at A and C are unknown. Each of these reactions are represented by two components: A_x and A_y at A , and C_x and C_y at C . The three equilibrium equations are.

$$+\curvearrowright \Sigma M_C = 0: \quad A_x(0.6 \text{ m}) - (30 \text{ kN})(0.8 \text{ m}) = 0$$

$$A_x = +40 \text{ kN} \quad (1.1)$$

$$+\rightarrow \Sigma F_x = 0: \quad A_x + C_x = 0$$

$$C_x = -A_x \quad C_x = -40 \text{ kN} \quad (1.2)$$

$$+\uparrow \Sigma F_y = 0: \quad A_y + C_y - 30 \text{ kN} = 0$$

$$A_y + C_y = +30 \text{ kN} \quad (1.3)$$

REVIEW OF THE METHODS OF STATICS

We have found two of the four unknowns, but cannot determine the other two from these equations, and no additional independent equation can be obtained from the free-body diagram of the structure. We must now dismember the structure. Considering the free-body diagram of the boom AB (Fig. 1.3), we write the following equilibrium equation:

$$+\circlearrowleft \Sigma M_B = 0: \quad -A_y(0.8 \text{ m}) = 0 \quad A_y = 0 \quad (1.4)$$

Substituting for A_y from Eq. (1.4) into Eq. (1.3), we obtain $C_y = +30 \text{ kN}$. Expressing the results obtained for the reactions at A and C in vector form, we have

$$A = 40 \text{ kN} \rightarrow \quad C_x = 40 \text{ kN} \leftarrow \quad C_y = 30 \text{ kN} \uparrow$$

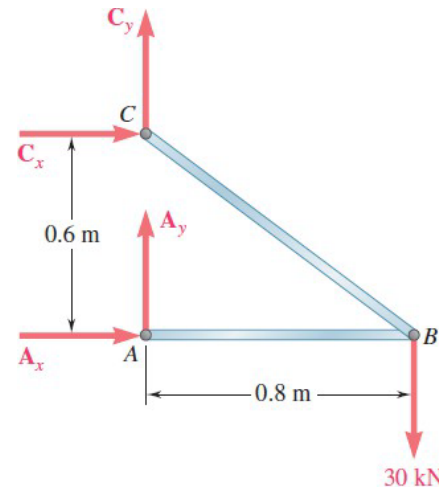


Fig. 1.2 Free-body diagram of boom showing applied load and reaction forces.

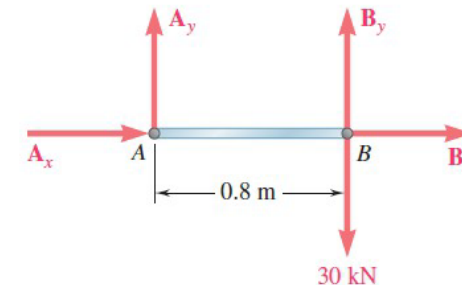
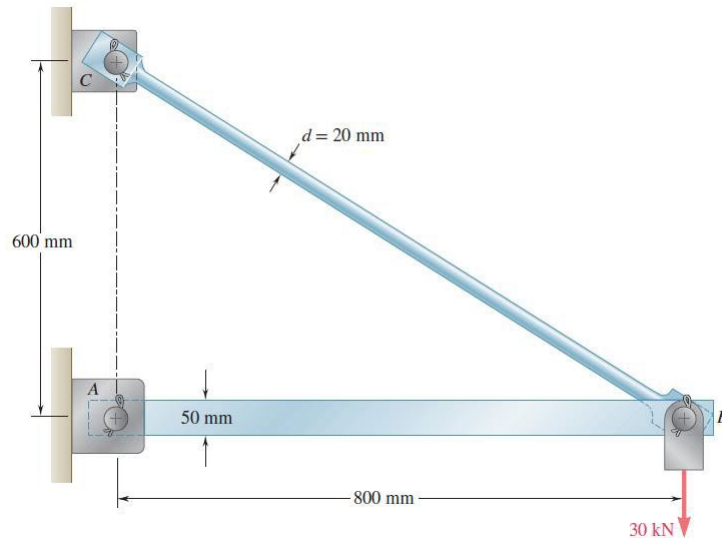


Fig. 1.3 Free-body diagram of member AB freed from structure.



REVIEW OF THE METHODS OF STATICS

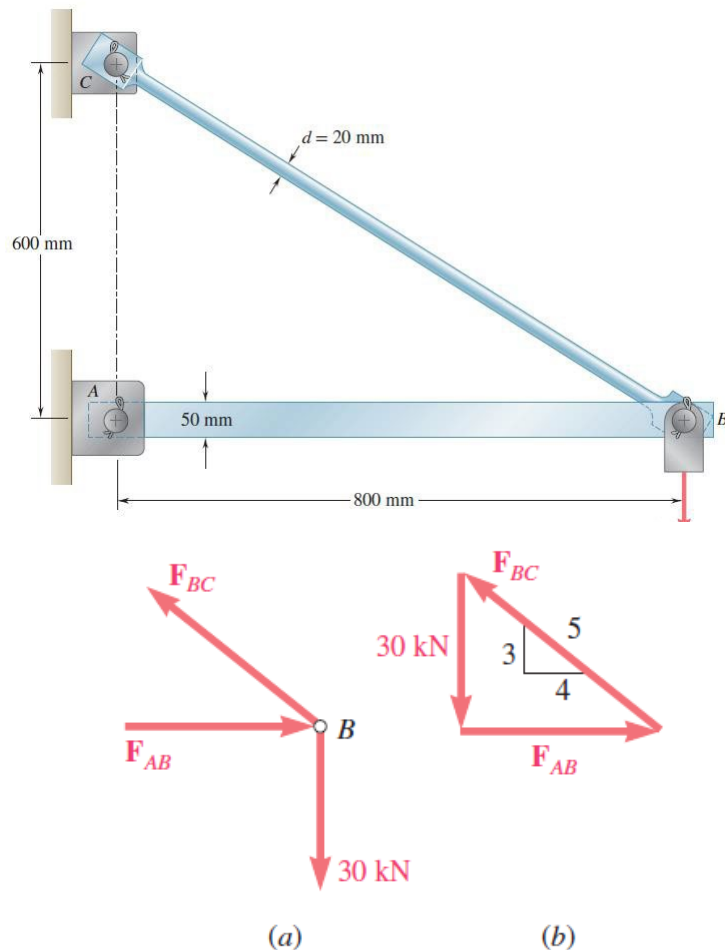


Fig. 1.4 Free-body diagram of boom's joint B and associated force triangle.

Note that the reaction at A is directed along the axis of the boom AB and causes compression in that member. Observe that the components C_x and C_y of the reaction at C are, respectively, proportional to the horizontal and vertical components of the distance from B to C and that the reaction at C is equal to 50 kN, is directed along the axis of the rod BC , and causes tension in that member.

These results could have been anticipated by recognizing that AB and BC are two-force members, i.e., members that are subjected to forces at only two points, these points being A and B for member AB , and B and C for member BC . Indeed, for a two-force member the lines of action of the resultants of the forces acting at each of the two points are equal and opposite and pass through both points. Using this property, we could have obtained a simpler solution by considering the free-body diagram of pin B . The forces on pin B , F_{AB} and F_{BC} , are exerted, respectively, by members AB and BC and the 30-kN load (Fig. 1.4a). Pin B is shown to be in equilibrium by drawing the corresponding force triangle (Fig. 1.4b).

Since force F_{BC} is directed along member BC , its slope is the same as that of BC , namely, $3/4$. We can, therefore, write the proportion

REVIEW OF THE METHODS OF STATICS

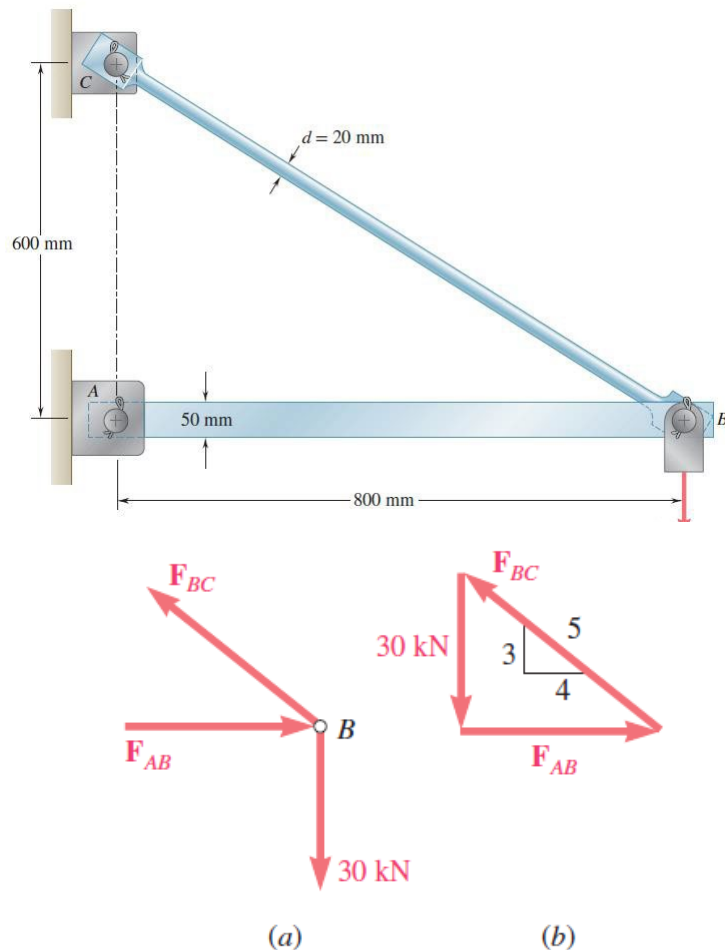


Fig. 1.4 Free-body diagram of boom's joint B and associated force triangle.

Note that the reaction at A is directed along the axis of the boom AB and causes compression in that member. Observe that the components C_x and C_y of the reaction at C are, respectively, proportional to the horizontal and vertical components of the distance from B to C and that the reaction at C is equal to 50 kN, is directed along the axis of the rod BC , and causes tension in that member.

These results could have been anticipated by recognizing that AB and BC are two-force members, i.e., members that are subjected to forces at only two points, these points being A and B for member AB , and B and C for member BC . Indeed, for a two-force member the lines of action of the resultants of the forces acting at each of the two points are equal and opposite and pass through both points. Using this property, we could have obtained a simpler solution by considering the free-body diagram of pin B . The forces on pin B , F_{AB} and F_{BC} , are exerted, respectively, by members AB and BC and the 30-kN load (Fig. 1.4a). Pin B is shown to be in equilibrium by drawing the corresponding force triangle (Fig. 1.4b).

Since force F_{BC} is directed along member BC , its slope is the same as that of BC , namely, $3/4$. We can, therefore, write the proportion

REVIEW OF THE METHODS OF STATICS

$$\frac{F_{AB}}{4} = \frac{F_{BC}}{5} = \frac{30 \text{ kN}}{3}$$

from which

$$F_{AB} = 40 \text{ kN} \quad F_{BC} = 50 \text{ kN}$$

Forces $\mathbf{F}'_{AB}$ and $\mathbf{F}'_{BC}$ exerted by pin B on boom AB and rod BC are equal and opposite to $\mathbf{F}_{AB}$ and $\mathbf{F}_{BC}$ (Fig. 1.5).

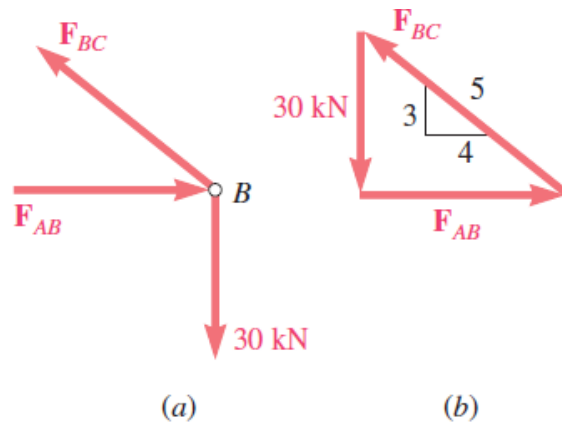


Fig. 1.4 Free-body diagram of boom's joint B and associated force triangle.

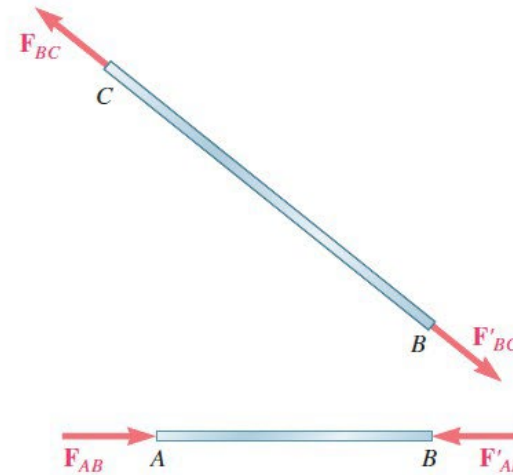


Fig. 1.5 Free-body diagrams of two-force members AB and BC .

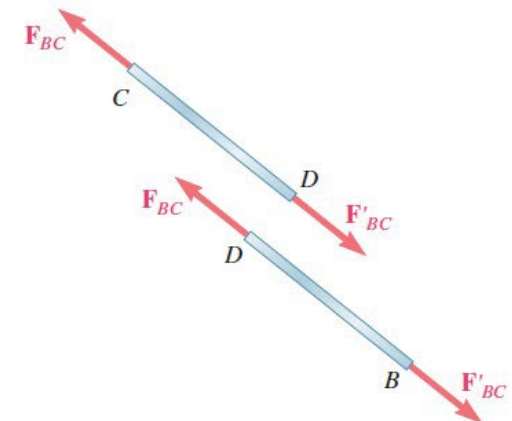
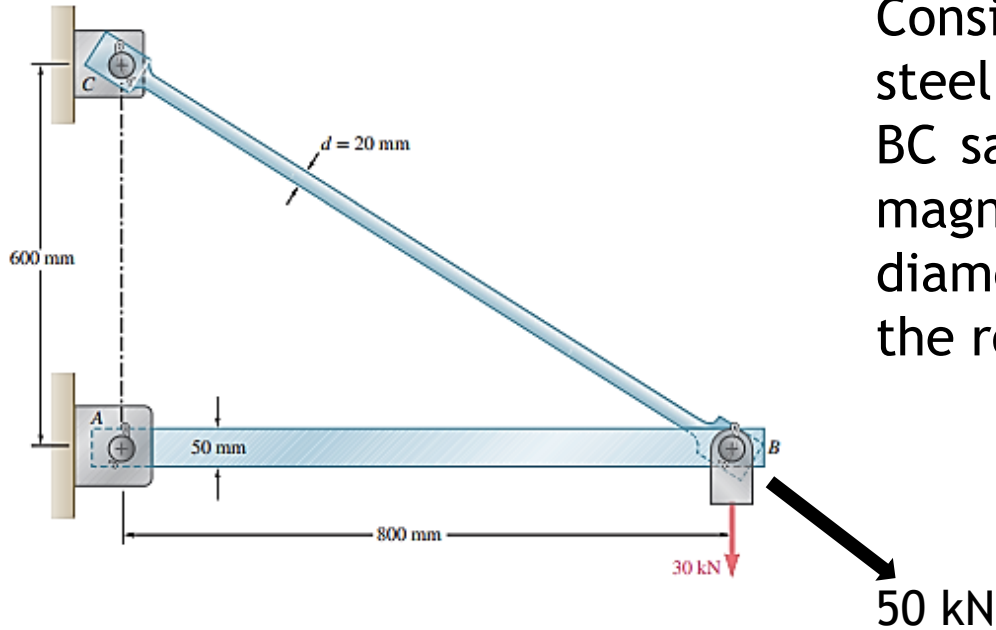


Fig. 1.6 Free-body diagrams of sections of rod BC .

Example:1



Considering the structure of , assume that rod BC is made of a steel with a maximum allowable stress $\sigma_{\text{all}} = 165 \text{ MPa}$. Can rod BC safely support the load to which it will be subjected? The magnitude of the force F_{BC} in the rod was 50 kN. If the diameter of the rod is 20 mm. Determine the stress created in the rod by the given loading.

$$P = F_{\text{BC}} = +50 \text{ kN} = +50 \times 10^3 \text{ N}$$

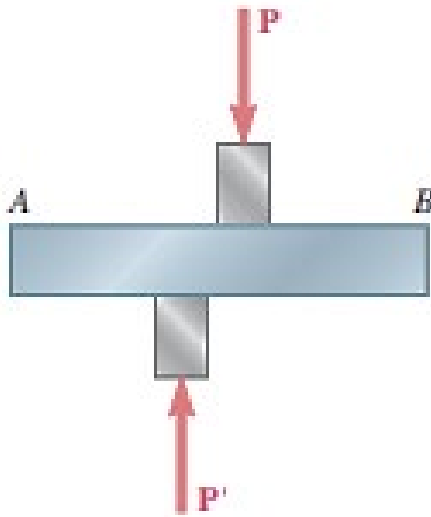
$$A = \pi r^2 = \pi \left(\frac{20 \text{ mm}}{2} \right)^2 = \pi (10 \times 10^{-3} \text{ m})^2 = 314 \times 10^{-6} \text{ m}^2$$

$$\sigma = \frac{P}{A} = \frac{+50 \times 10^3 \text{ N}}{314 \times 10^{-6} \text{ m}^2} = +159 \times 10^6 \text{ Pa} = +159 \text{ MPa}$$

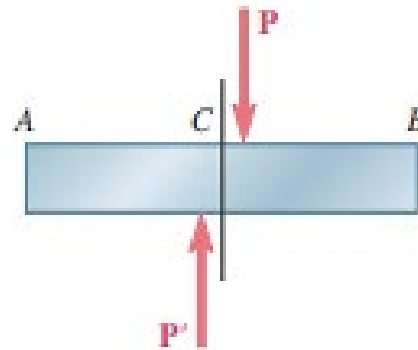
Since σ is smaller than σ_{all} of the allowable stress in the steel used, rod BC can safely support the load.

Shearing Stress

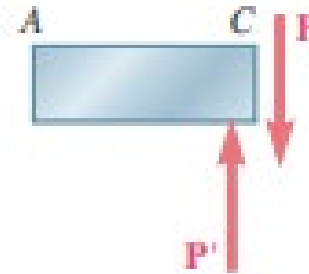
- Forces P and P' are applied transversely to the member AB .
- Corresponding internal forces act in the plane of section C are called **shearing forces**.



Opposing transverse loads
creating shear on member AB



The resulting internal shear force on a section between
transverse forces.

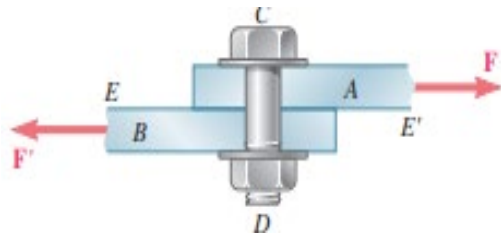


Shearing Stress Examples

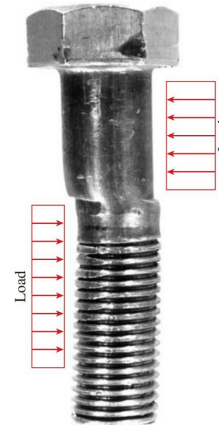
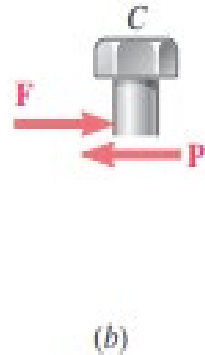
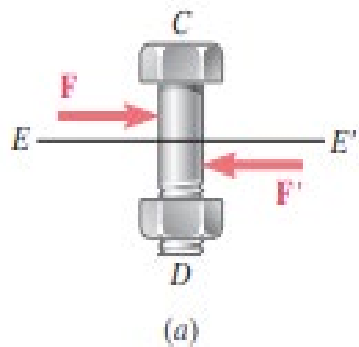
Dividing the shear **P** by the area **A** of the cross section, you obtain the average shearing stress in the section. Denoting the shearing stress by the Greek letter τ (tau)

$$\tau_{ave} = \frac{P}{A}$$

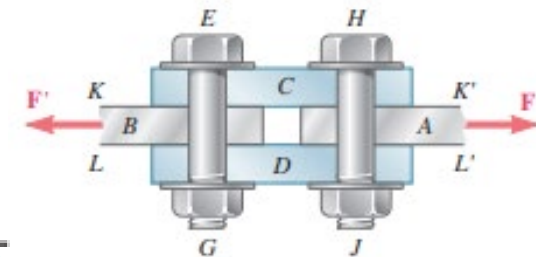
Single Shear



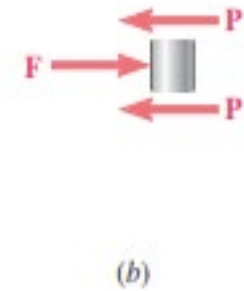
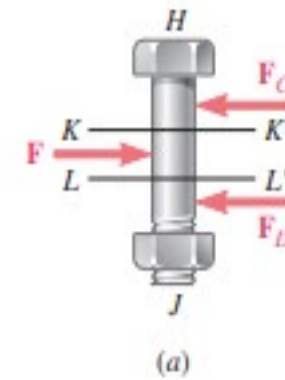
$$\tau_{ave} = \frac{P}{A} = \frac{F}{A}$$



Double Shear

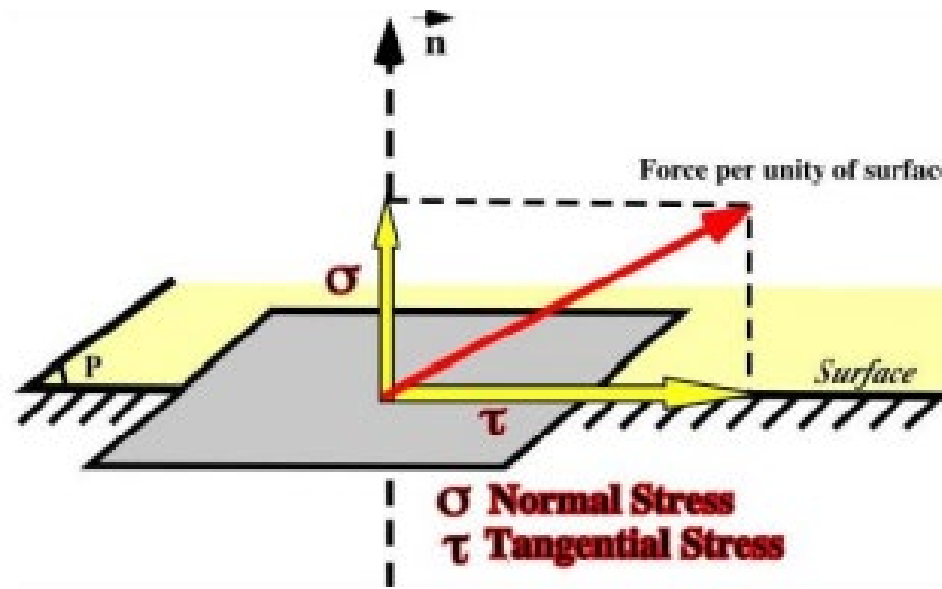


$$\tau_{ave} = \frac{P}{A} = \frac{F/2}{A} = \frac{F}{2A}$$

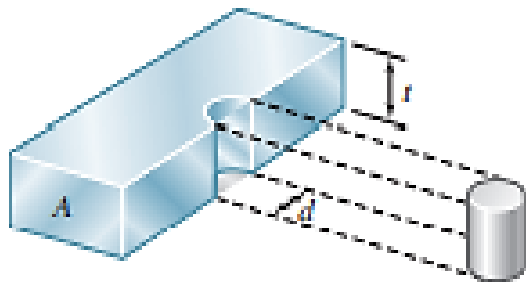
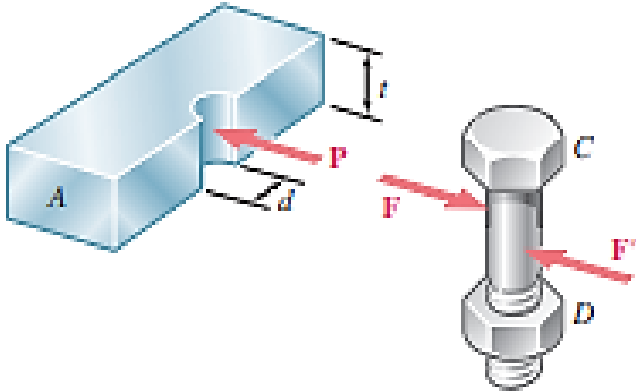


Two Components of stress

- The stress (resulting stress) at any point in a body is basically resolved into two components σ and τ .
- One acts perpendicular and other parallel to the area concerned, as it is clearly defined in the following figure.



Bearing Stress in Connections

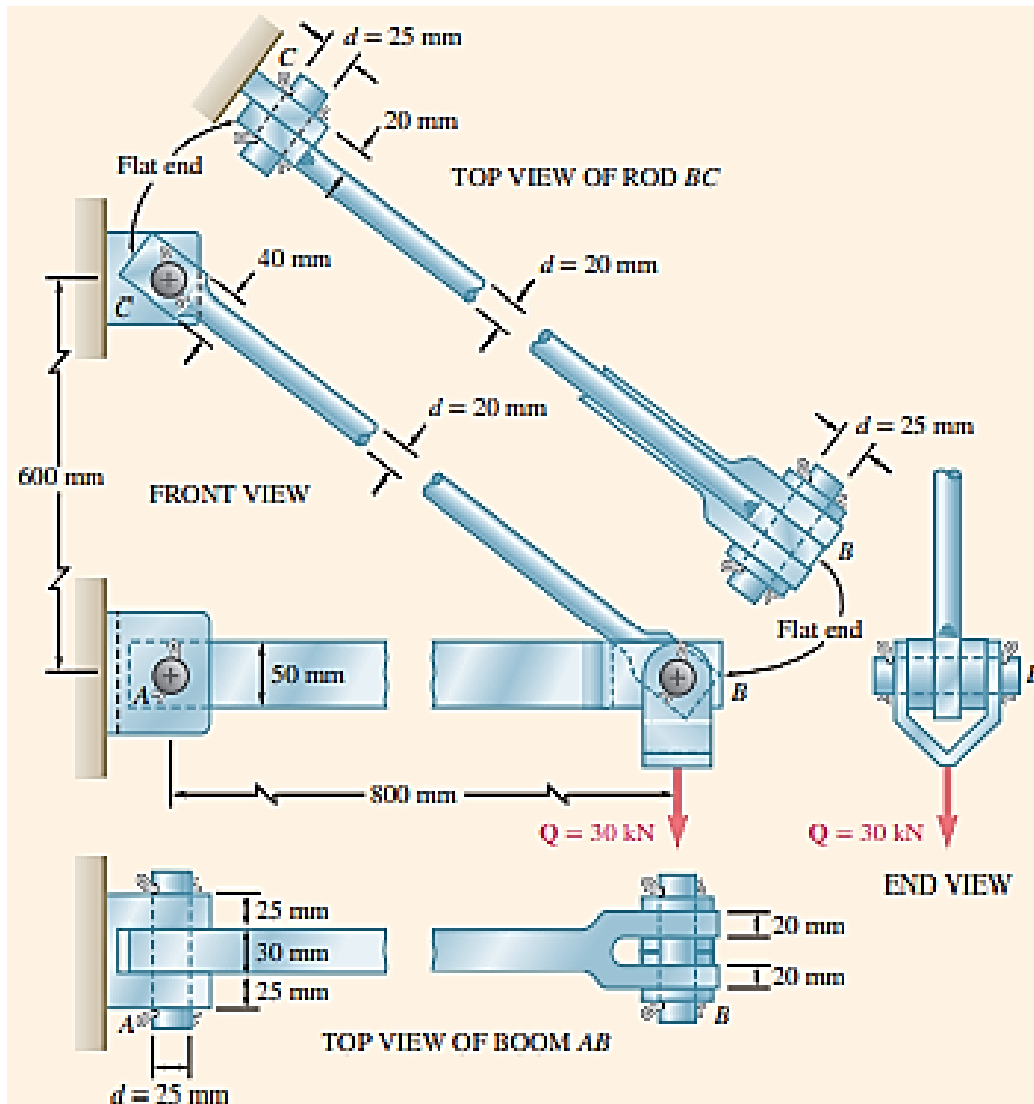


Dimensions for calculating bearing stress area.

- Bolts, rivets, and pins create stresses on the points of contact or *bearing surfaces* of the members they connect.
- The resultant of the *force distribution* on the surface is equal and opposite to the force exerted on the pin.
- Corresponding average force intensity is called the bearing stress.

$$\sigma_b = \frac{P}{A} = \frac{P}{td}$$

Concept Application 1.3: Stress Analysis & Design Example



From Static Analysis
 $F_{AB} = 40$ kN (Compression)
 $F_{BC} = 50$ kN (Tension)

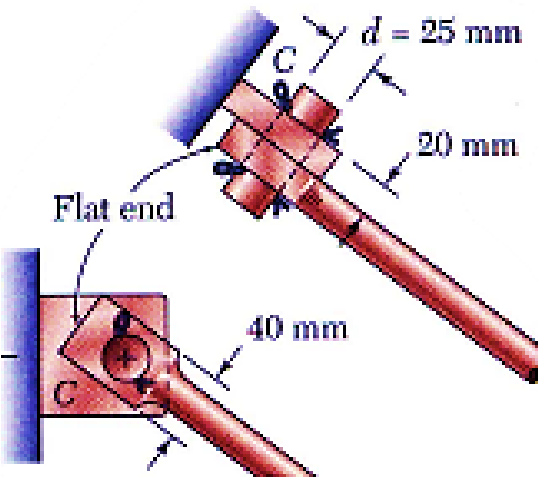
- Determine the stresses in the members and connections of the structure shown.
- 20-mm diameter rod BC has flat ends of 20×40 mm rectangular cross section,
- While boom AB has a 30×50 mm rectangular cross section and is fitted with a clevis at end B.
- Both members are connected at B by a pin from which the 30-kN load is suspended by means of a U-shaped bracket.
- Boom AB is supported at A by a pin fitted into a double bracket, while rod BC is connected at C to a single bracket. All pins are 25 mm in diameter

The force in rod BC is $F_{BC} = 50$ kN (tension) and the area of its circular cross section is $A = 314 \times 10^{-6} \text{ m}^2$

Normal Stress in Boom AB and Rod BC.

- force in rod BC is $F_{BC} = 50 \text{ kN}$ (tension)
- area of its circular cross section is $A = 314 \times 10^{-6} \text{ m}^2$.

The corresponding average normal stress is $\sigma_{BC} = +159 \text{ MPa}$. However, the flat parts of the rod are also under tension and at the narrowest section. Where the hole is located, we have $A = (20 \text{ mm})(40 \text{ mm} - 25 \text{ mm}) = 300 \times 10^{-6} \text{ m}^2$



$$\sigma_{BC,end} = \frac{P}{A} = \frac{50 \times 10^3 \text{ N}}{300 \times 10^{-6} \text{ m}^2} = 167 \text{ MPa}$$

- The boom is in compression with an axial force of 40kN and average normal stress of -26.7 MPa .
- The minimum area sections at the boom ends are unstressed since the boom is in compression.

Shear Stress in Pins

- The cross-sectional area for pins at A , B , and C ,

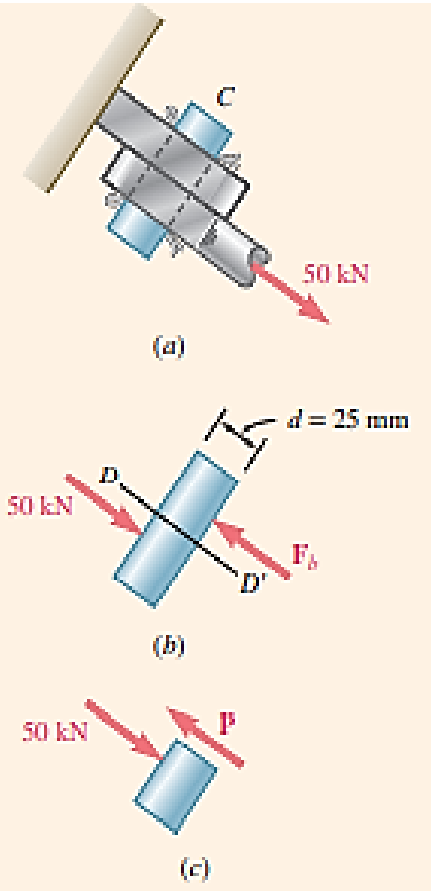
$$A = \pi r^2 = \pi \left(\frac{25 \text{ mm}}{2} \right)^2 = 491 \times 10^{-6} \text{ m}^2$$

- The force on the pin at C is equal to the force exerted by the rod BC ,

$$\tau_{C,ave} = \frac{P}{A} = \frac{50 \times 10^3 \text{ N}}{491 \times 10^{-6} \text{ m}^2} = 102 \text{ MPa}$$

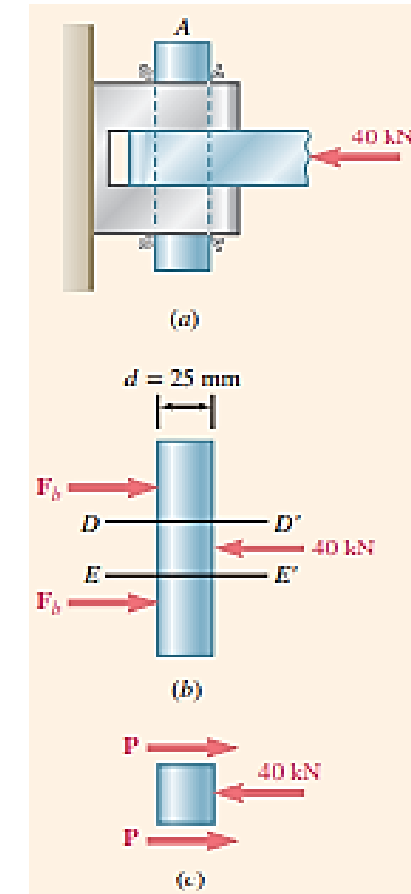
- The pin at A is in double shear with a total force equal to the force exerted by the boom AB ,

$$\tau_{A,ave} = \frac{P}{A} = \frac{20 \text{ kN}}{491 \times 10^{-6} \text{ m}^2} = 40.7 \text{ MPa}$$

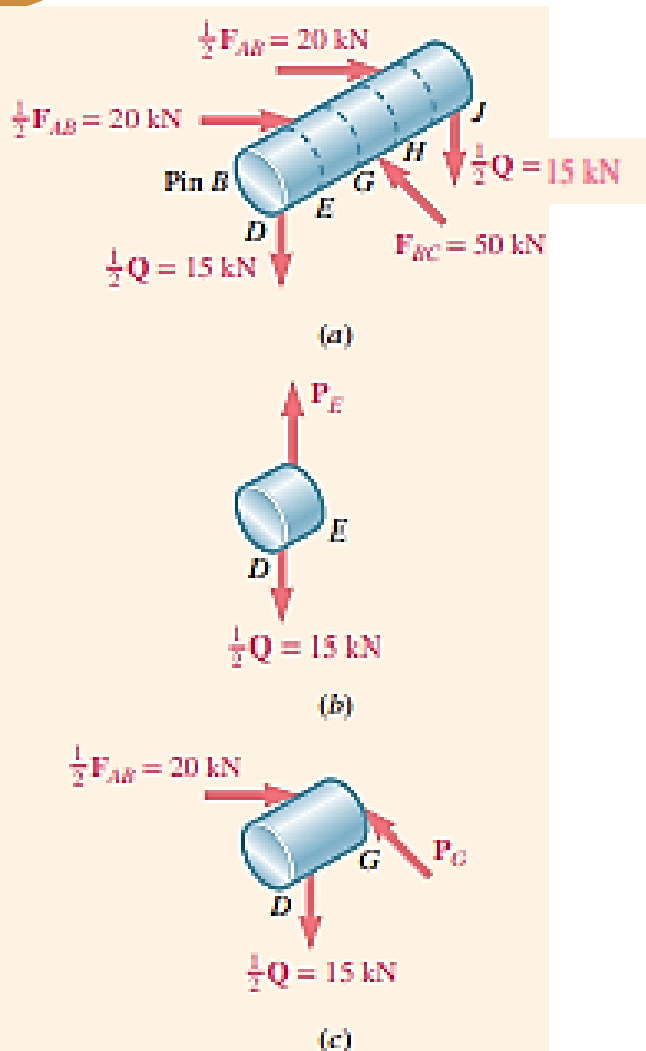


Diagrams of the single-shear pin at C .

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Free-body diagrams of the double-shear pin at A



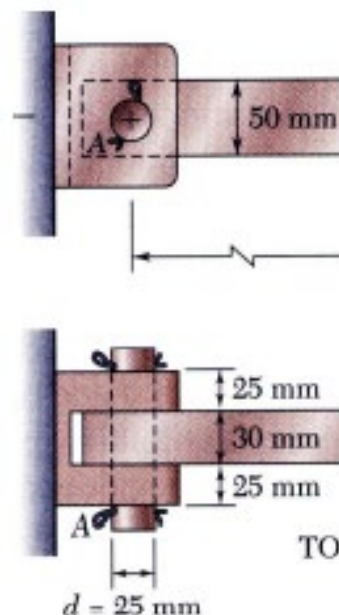
- Divide the pin at B into sections to determine the section with the largest shear force,

$$P_E = 15 \text{ kN}$$

$$P_G = 25 \text{ kN (largest)}$$

- Evaluate the corresponding average shearing stress,

$$\tau_{B,ave} = \frac{P_G}{A} = \frac{25 \text{ kN}}{491 \times 10^{-6} \text{ m}^2} = 50.9 \text{ MPa}$$



- To determine the bearing stress at A in the boom AB , we have $t = 30 \text{ mm}$ and $d = 25 \text{ mm}$,

$$\sigma_b = \frac{P}{td} = \frac{40 \text{ kN}}{(30 \text{ mm})(25 \text{ mm})} = 53.3 \text{ MPa}$$

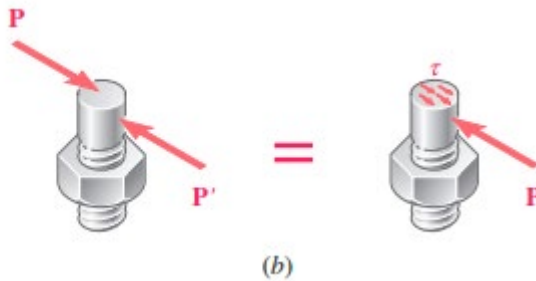
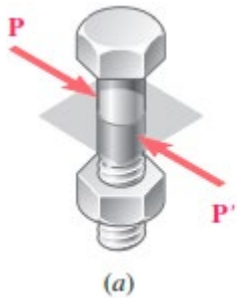
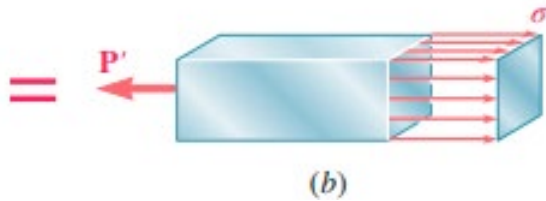
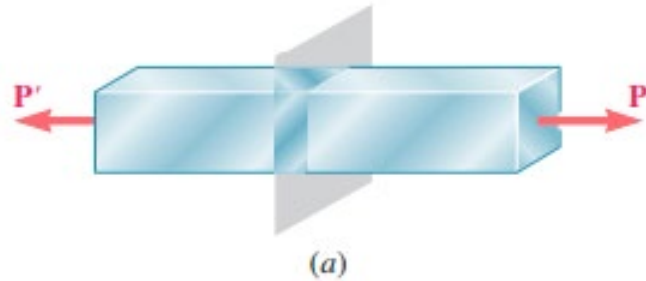
- To determine the bearing stress at A in the bracket, we have $t = 2(25 \text{ mm}) = 50 \text{ mm}$ and $d = 25 \text{ mm}$,

$$\sigma_b = \frac{P}{td} = \frac{40 \text{ kN}}{(50 \text{ mm})(25 \text{ mm})} = 32.0 \text{ MPa}$$



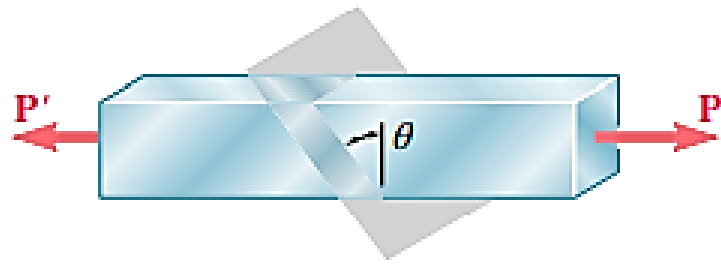
Axial Loaded Members

Stress in Two Force Members



- Axial forces on a two force member result in only normal stresses on a plane cut perpendicular to the member axis.
- Transverse forces on bolts and pins result in only shear stresses on the plane perpendicular to bolt or pin axis.
- Will show that either axial or transverse forces may produce both normal and shear stresses with respect to a plane other than one cut perpendicular to the member axis.

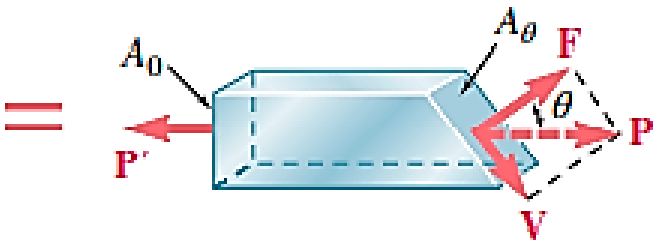
Stress on an Oblique Plane



(a)



(b)



(c)



(d)

- Pass a section through the member forming an angle θ with the normal plane.

- From equilibrium conditions, the distributed forces (stresses) on the plane must be equivalent to the force P
- Resolve P into components normal and tangential to the oblique section,

$$F = P \cos \theta \quad V = P \sin \theta$$

- The average normal and shear stresses on the oblique plane are

$$\sigma = \frac{F}{A_\theta} = \frac{P \cos \theta}{\frac{A_0}{\cos \theta}} = \frac{P}{A_0} \cos^2 \theta$$

$$\tau = \frac{V}{A_\theta} = \frac{P \sin \theta}{\frac{A_0}{\cos \theta}} = \frac{P}{A_0} \sin \theta \cos \theta$$

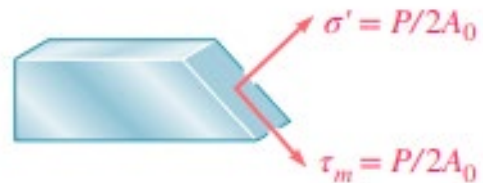
Maximum Stresses



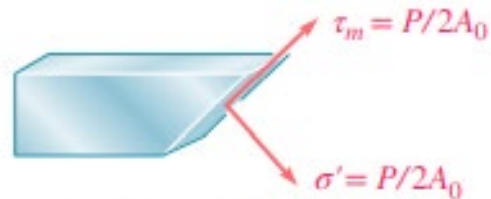
(a) Axial loading



(b) Stresses for $\theta = 0$



(c) Stresses for $\theta = 45^\circ$



(d) Stresses for $\theta = -45^\circ$

- The maximum normal stress occurs when the reference plane is perpendicular to the member axis,

$$\sigma = \frac{P}{A_0} \cos^2 \theta \quad \tau = \frac{P}{A_0} \sin \theta \cos \theta$$

- The maximum shear stress occurs for a plane at $\pm 45^\circ$ with respect to the axis,

$$\tau_m = \frac{P}{A_0} \sin 45 \cos 45 = \frac{P}{2A_0} = \sigma'$$

Stress Under General Loadings

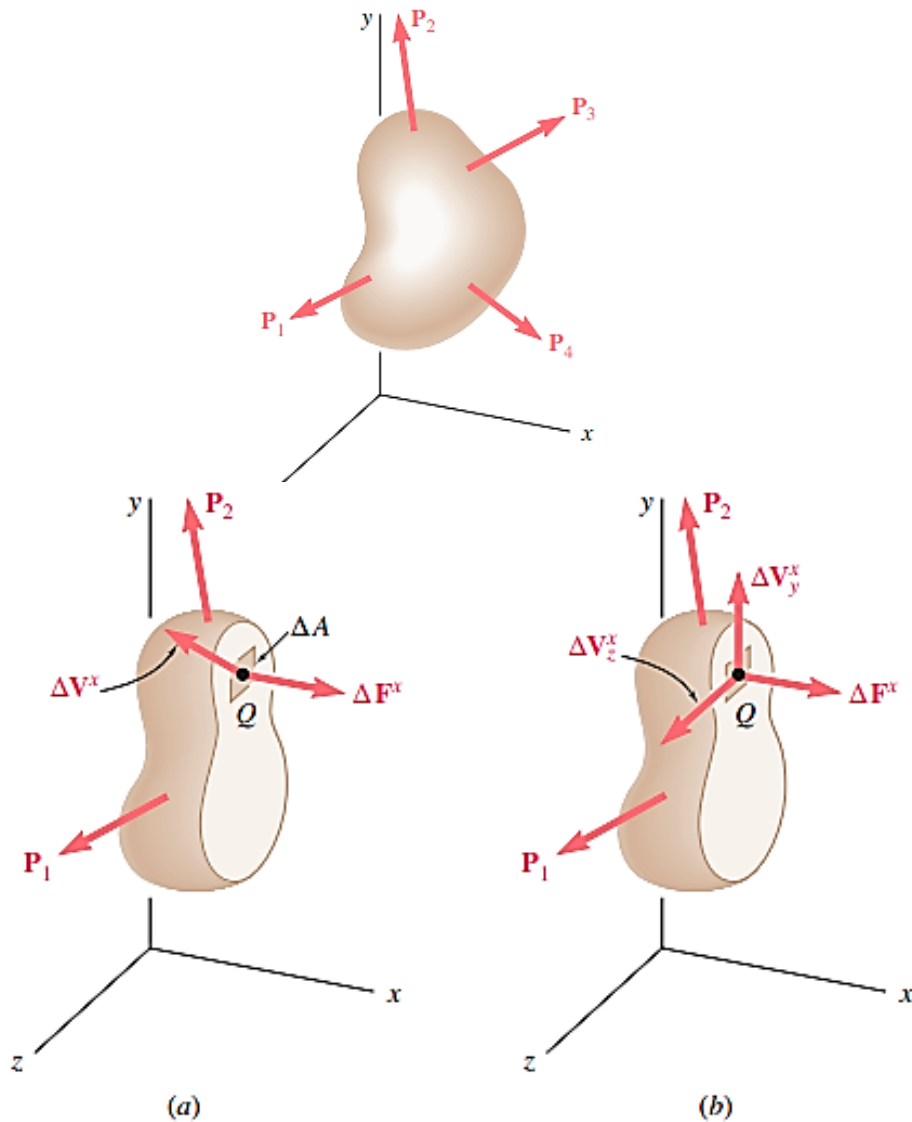
A member subjected to a general combination of loads is cut into two segments by a plane passing through Q.

The distribution of internal stress components may be defined as,

$$\sigma_x = \lim_{\Delta A \rightarrow 0} \frac{\Delta F^x}{\Delta A}$$

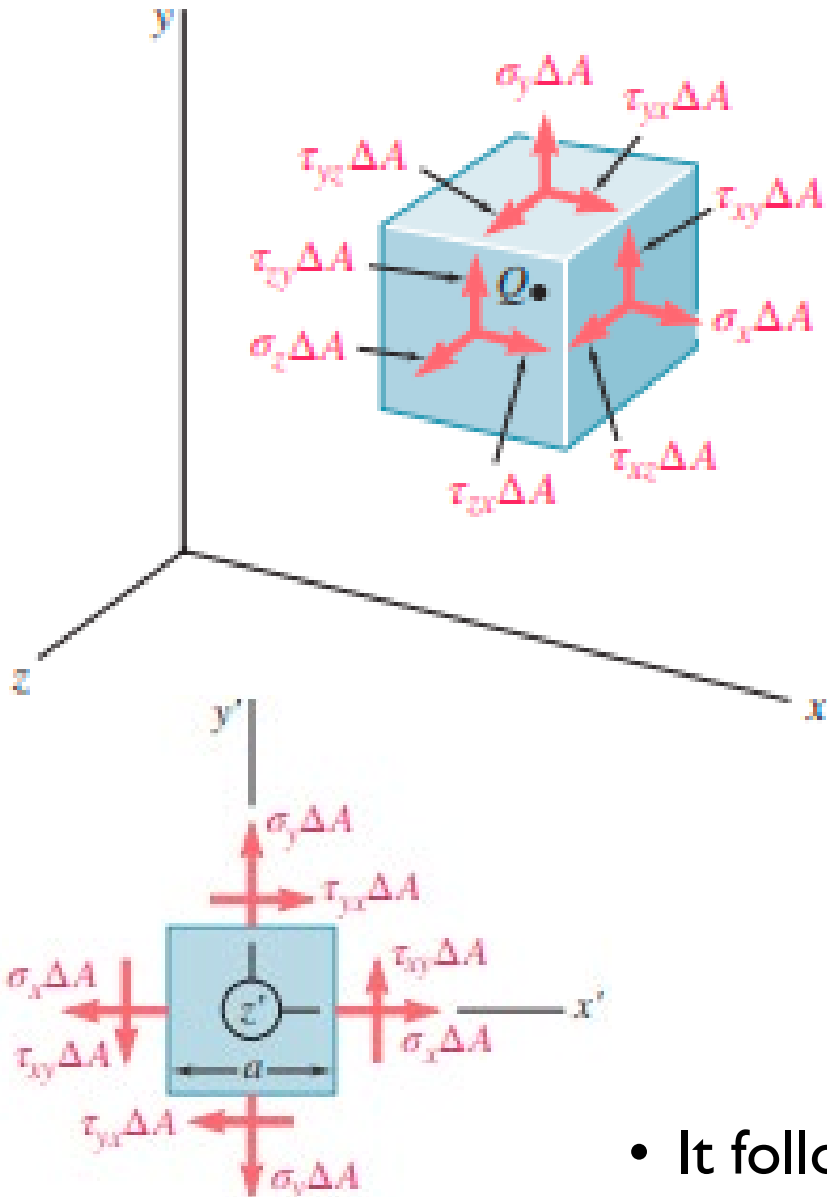
$$\tau_{xy} = \lim_{\Delta A \rightarrow 0} \frac{\Delta V_y^x}{\Delta A}$$

$$\tau_{xz} = \lim_{\Delta A \rightarrow 0} \frac{\Delta V_z^x}{\Delta A}$$



For equilibrium, an equal and opposite internal force and stress distribution must be exerted on the other segment of the member

State of Stress



- Stress components are defined for the planes cut parallel to the x, y and z axes. For equilibrium, equal and opposite stresses are exerted on the hidden planes.

- The combination of forces generated by the stresses must satisfy the conditions for equilibrium:

$$\sum F_x = \sum F_y = \sum F_z = 0$$

$$\sum M_x = \sum M_y = \sum M_z = 0$$

- Consider the moments about the z axis:

$$\sum M_z = 0 = (\tau_{xy}\Delta A)a - (\tau_{yx}\Delta A)a$$

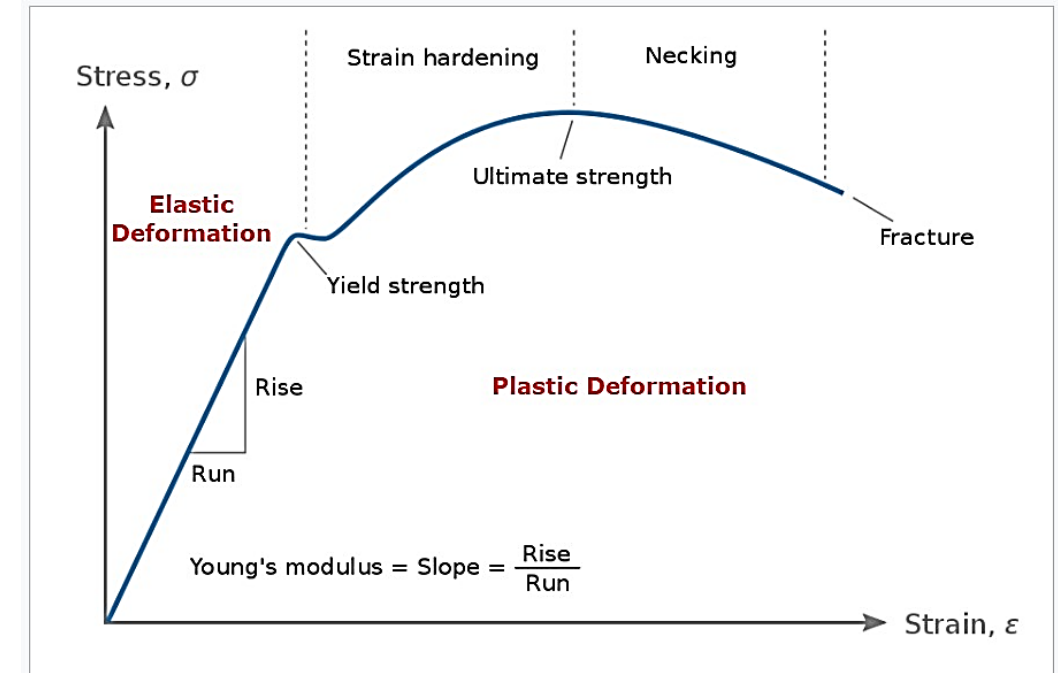
$$\tau_{xy} = \tau_{yx} \quad \text{similarly,} \quad \tau_{yz} = \tau_{zy} \quad \text{and} \quad \tau_{xz} = \tau_{zx}$$

- It follows that only 6 components of stress are required to define the complete state of stress

Design Consideration

A. Determination of Ultimate Strength of a Material

B. Allowable Load and Allowable Stress:



Factor of Safety

Structural members or machines must be designed such that the working stresses are less than the ultimate strength of the material.

FS = Factor of safety

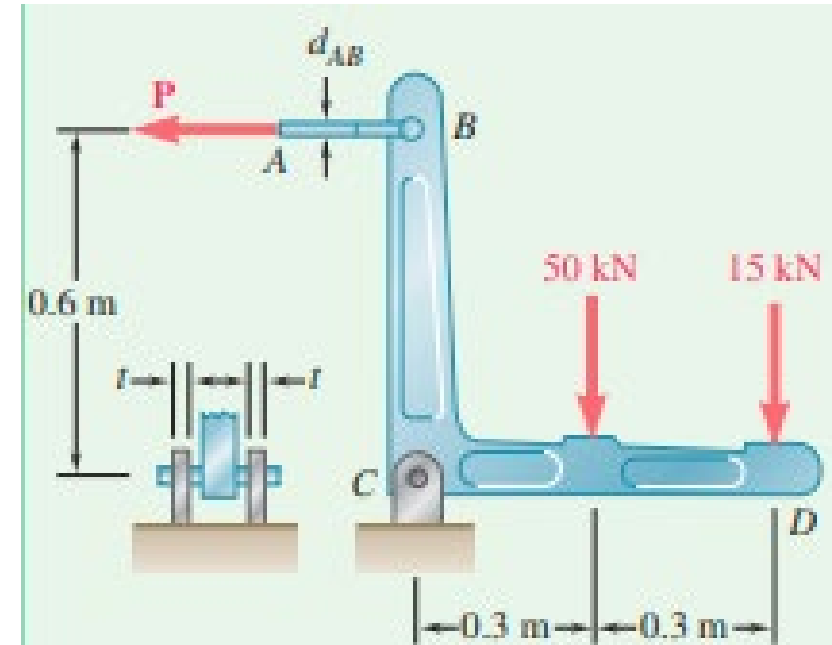
$$FS = \frac{\sigma_u}{\sigma_{all}} = \frac{\text{Ultimate stress}}{\text{Allowable stress}}$$

Factor of safety considerations:

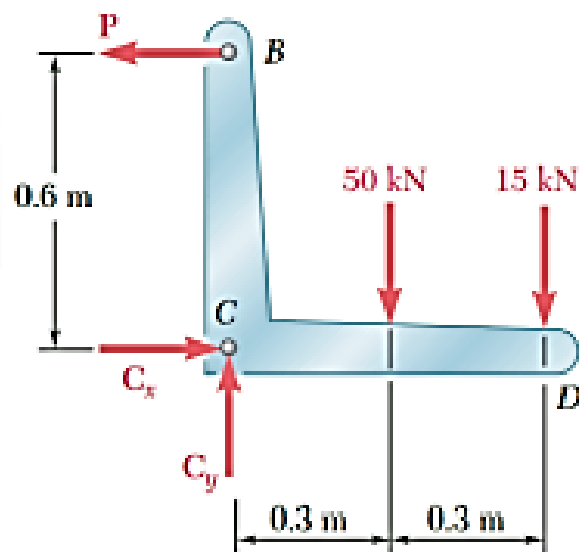
- uncertainty in material properties
- uncertainty of loadings
- uncertainty of analyses
- number of loading cycles
- types of failure
- maintenance requirements and deterioration effects
- risk to life and property
- influence on machine function

Sample Problem 1.3

Two loads are applied to the bracket BCD as shown. (a) Knowing that the control rod AB is to be made of a steel having an ultimate normal stress of 600 MPa, determine the diameter of the rod for which the factor of safety with respect to failure will be 3.3. (b) The pin at C is to be made of a steel having an ultimate shearing stress of 350 MPa. Determine the diameter of the pin C for which the factor of safety with respect to shear will also be 3.3. (c) Determine the required thickness of the bracket supports at C , knowing that the allowable bearing stress of the steel used is 300 MPa



Solution



$$\sum M_C = 0: P(0.6) - (50 \times 0.3) - (15 \times 0.6) = 0$$

$$P = 40 \text{ kN}$$

$$\sum F_x = 0: C_x = 40 \text{ kN}$$

$$\sum F_y = 0: C_y = 65 \text{ kN} \quad C = \sqrt{C_x^2 + C_y^2} = 76.3 \text{ kN}$$

a. Control Rod AB. $\sigma_{all} = \frac{\sigma_U}{F.S.} = \frac{600}{3.3} = 181.8 \text{ MPa}$

Since $\sigma_{all} = 181.8 \times 10^6 = \frac{P}{A_{AB}} = \frac{40 \times 10^3}{\frac{\pi}{4} d_{AB}^2}$

$$d_{AB} = 16.74 \times 10^{-3} \text{ m} = 16.74 \text{ mm}$$

b. Shear in Pin C. $\tau_{all} = \frac{\tau_U}{F.S.} = \frac{350}{3.3} = 106.1 \text{ MPa}$

As shown in Fig. 2 the pin is in double shear.

$$\tau_{all} = 106.1 \times 10^6 = \frac{C}{2A_C} = \frac{76.3 \times 10^3}{2 \times \frac{\pi}{4} d_C^2}$$

$$d_C = 21.4 \times 10^{-3} \text{ m} = 21.4 \text{ mm}$$

$$\text{Use } d_C = 22 \text{ mm}$$

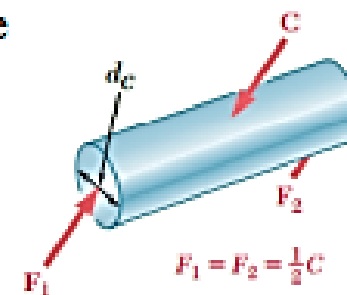


Fig. 2 Free-body diagram of pin at point C.

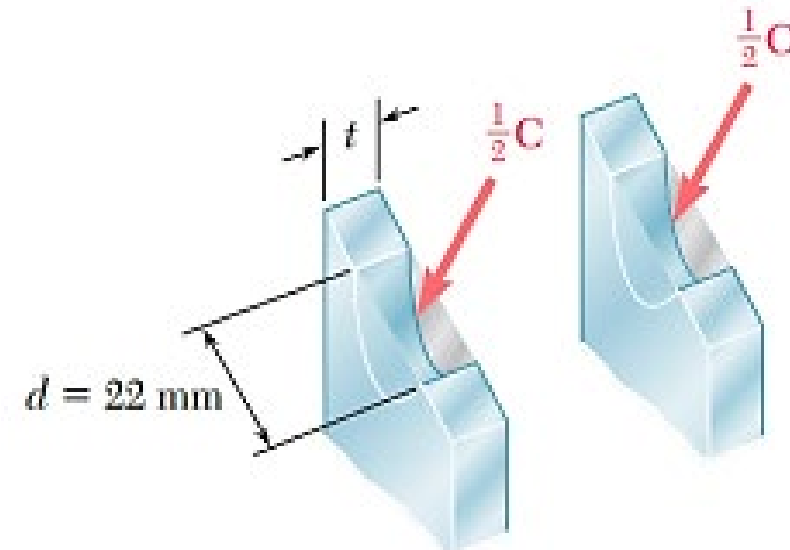
c. Bearing at C.

The force carried by each bracket is $C/2$ and allowable bearing stress is 300 MPa

$$\sigma_b = \frac{C}{2A} = \frac{C}{2td_c}$$

$$t = \frac{C}{2\sigma_b d_c} = \frac{76.3 \times 10^3}{2 \times 300 \times 22} = 5.78 \text{ mm}$$

Use $t = 6 \text{ mm}$





Strain

This sub-chapter considers deformations occurring in structural components subjected to axial loading.

The change in length of the diagonal stays was carefully accounted for in the design of this cable-stayed bridge.

AN INTRODUCTION TO STRESS AND STRAIN

Normal Strain Under Axial Loading

If the bar is subjected to a direct load, due to stress in the bar, there will be a change in length. If the bar has an original length “ L ” and change by an amount “ δ ”, strain (ϵ) is,

$$\epsilon = \frac{\delta}{L}$$

Sign Conventions

- Tensile Strain = +ve
- Compressive Strain = -ve

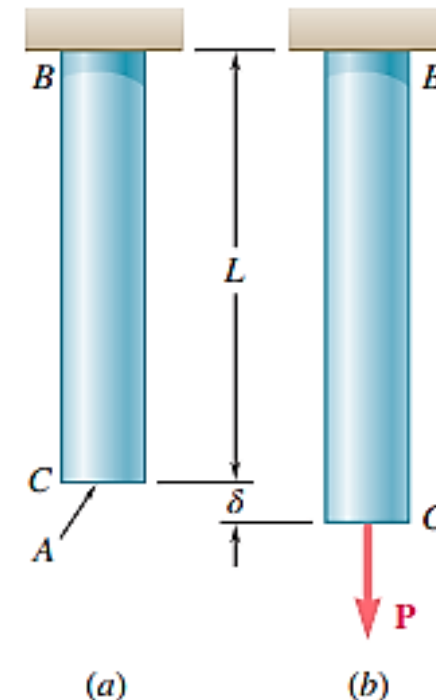
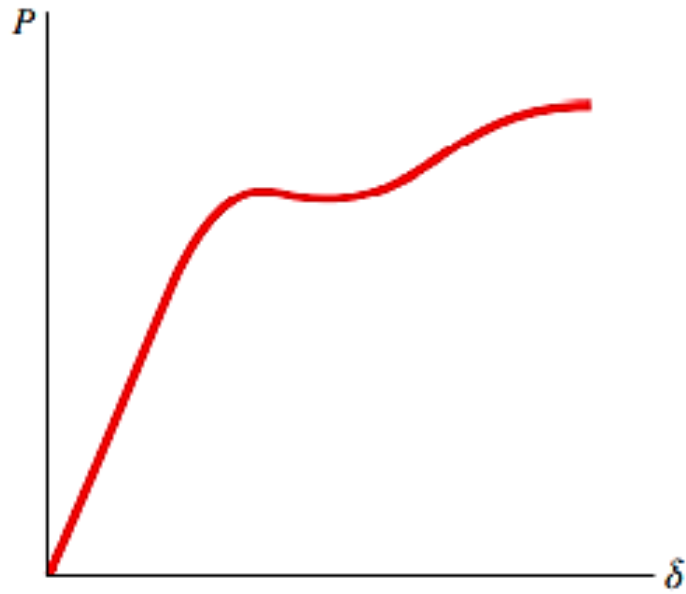
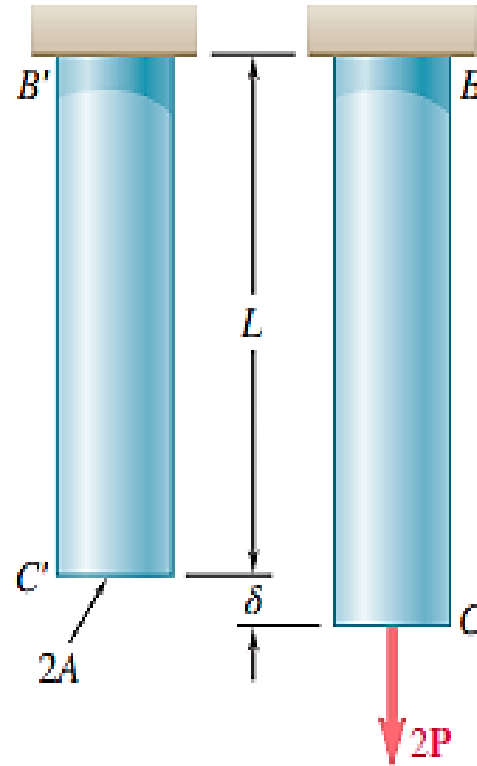


Fig. 2.1 Undeformed and deformed axially loaded rod.

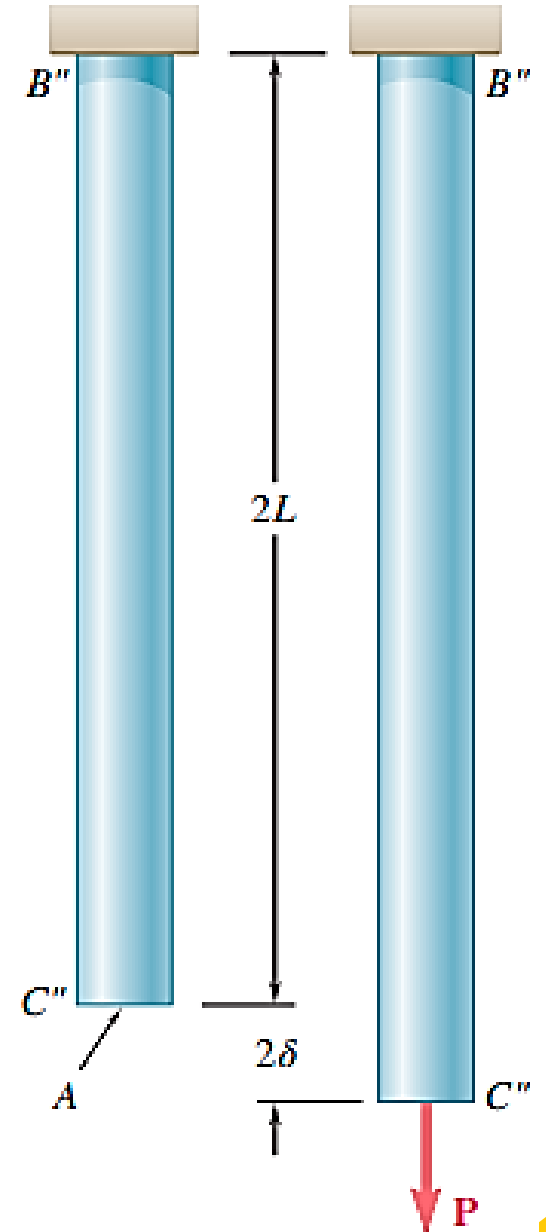


stress $\sigma = P/A$ against the strain $\varepsilon = \delta/L$



$$\sigma = \frac{2P}{2A} = \frac{P}{A}$$

$$\varepsilon = \frac{\delta}{L}$$



$$\sigma = \frac{P}{A}$$

$$\varepsilon = \frac{2\delta}{2L} = \frac{\delta}{L}$$

Hook's Law

- All deformable materials deform under loads
- A material is said to be elastic if it returns to its original position after removal of load.

In the elastic materials, stress is proportional to strain.
(Hook's Law)

$$\sigma \propto \varepsilon$$

$$\sigma = E\varepsilon$$

$$E = \frac{\sigma}{\varepsilon} \rightarrow \text{Modulus of elasticity}$$

Valid in the elastic region

Stress and Strain Test



Mechanical Properties of Materials

Introduction

- The design of machines and structures so that they will function properly requires that we understand the mechanical behavior of the materials being used.
- Ordinarily, the only way to determine how materials behave when they are subjected to loads is to perform experiments in the laboratory.
- The usual procedure is to place small specimens of the material in testing machines, apply the loads, and then measure the resulting deformations (such as changes in length and changes in diameter).
- Most materials-testing laboratories are equipped with machines capable of loading specimens in a variety of ways, including both static and dynamic loading in tension and compression.
- In a properly designed specimen, failure will occur in the prismatic portion of the specimen where the stress distribution is uniform and the bar is subjected only to pure tension.

A typical tensile-test machine

Tensile-test machine with automatic data-processing system.



A typical compression-test machine



A typical compression-test machine



Standardization

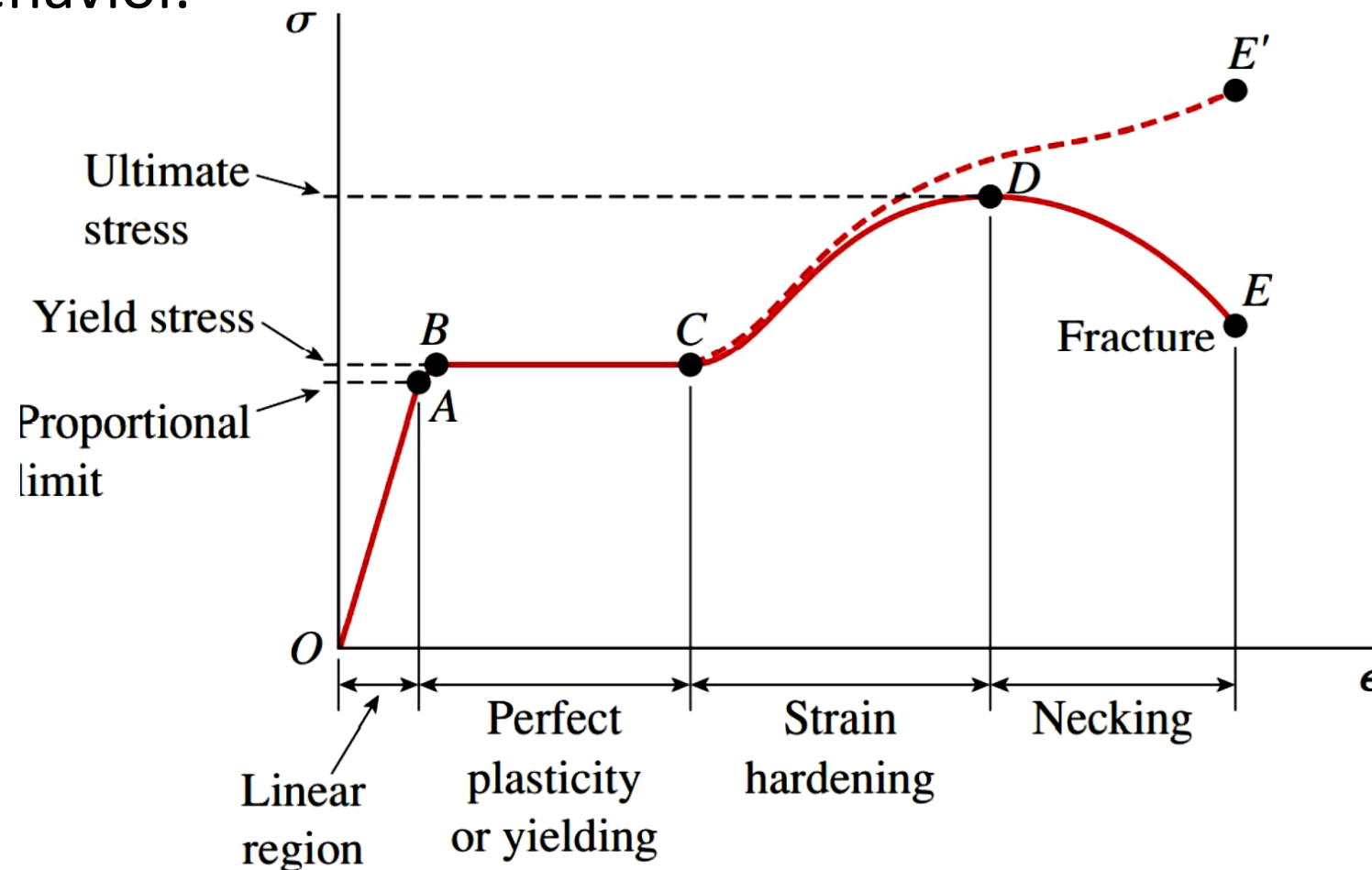
- In a properly designed specimen, failure will occur in the prismatic portion of the specimen where the stress distribution is uniform and the bar is subjected only to pure tension.
- In order that test results will be comparable, the dimensions of test specimens and the methods of applying loads must be standardized.
- One of the major standards organizations in the United States is the American Society for Testing and Materials (ASTM), a technical society that publishes specifications and standards for materials and testing.

Static and Dynamic Tests

- In a static test, the load is applied slowly and the precise rate of loading is not of interest because it does not affect the behavior of the specimen.
- However, in a dynamic test the load is applied rapidly and sometimes in a cyclical manner. Since the nature of a dynamic load affects the properties of the materials, the rate of loading must also be measured.

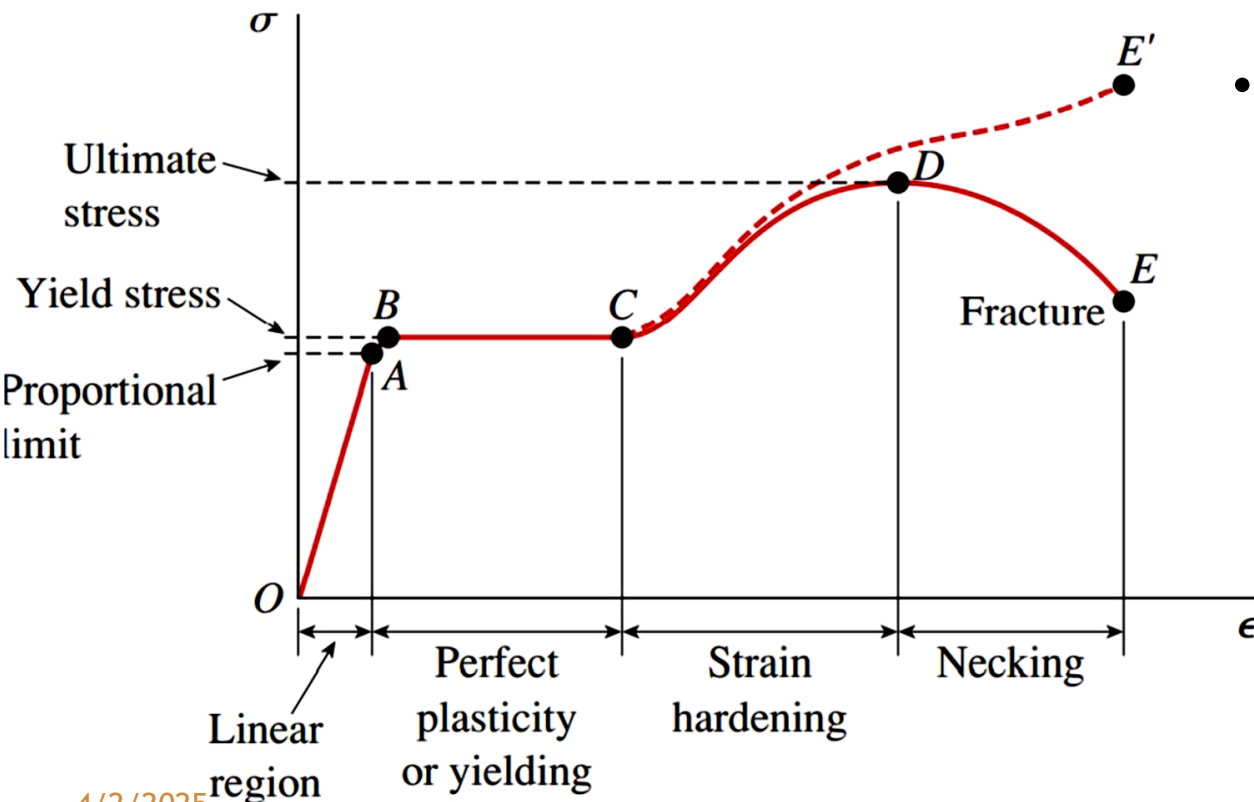
Stress-strain diagram for a typical structural steel in tension

Stress-strain diagram is a characteristic of the particular material being tested and conveys important information about the mechanical properties and type of behavior.



The Proportional limit

- Beyond point A, the proportionality between stress and strain no longer exists; hence the stress at A is called the proportional limit.
- The slope of the straight line from O to A is called the **modulus of elasticity**. Because the slope has units of stress divided by strain, modulus of elasticity has the same units as stress.

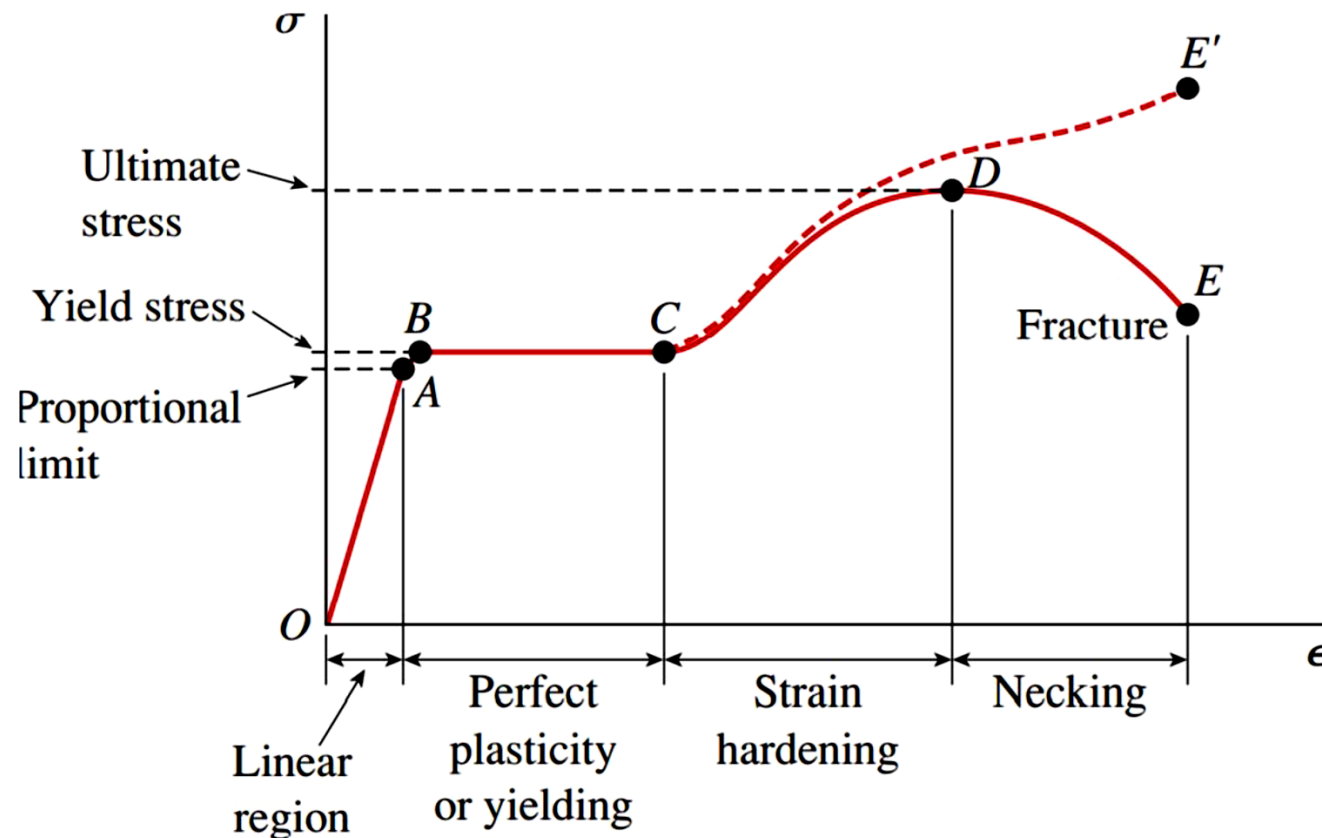


- At point B, considerable elongation with no noticeable increase in the tensile force (from B to C). This phenomenon is known as **yielding** of the material, and point B is called the **yield point**. The corresponding stress is known as the **yield stress** of the steel.

In the region from B to C, the material becomes **perfectly plastic**, which means that it deforms without an increase in the applied load.

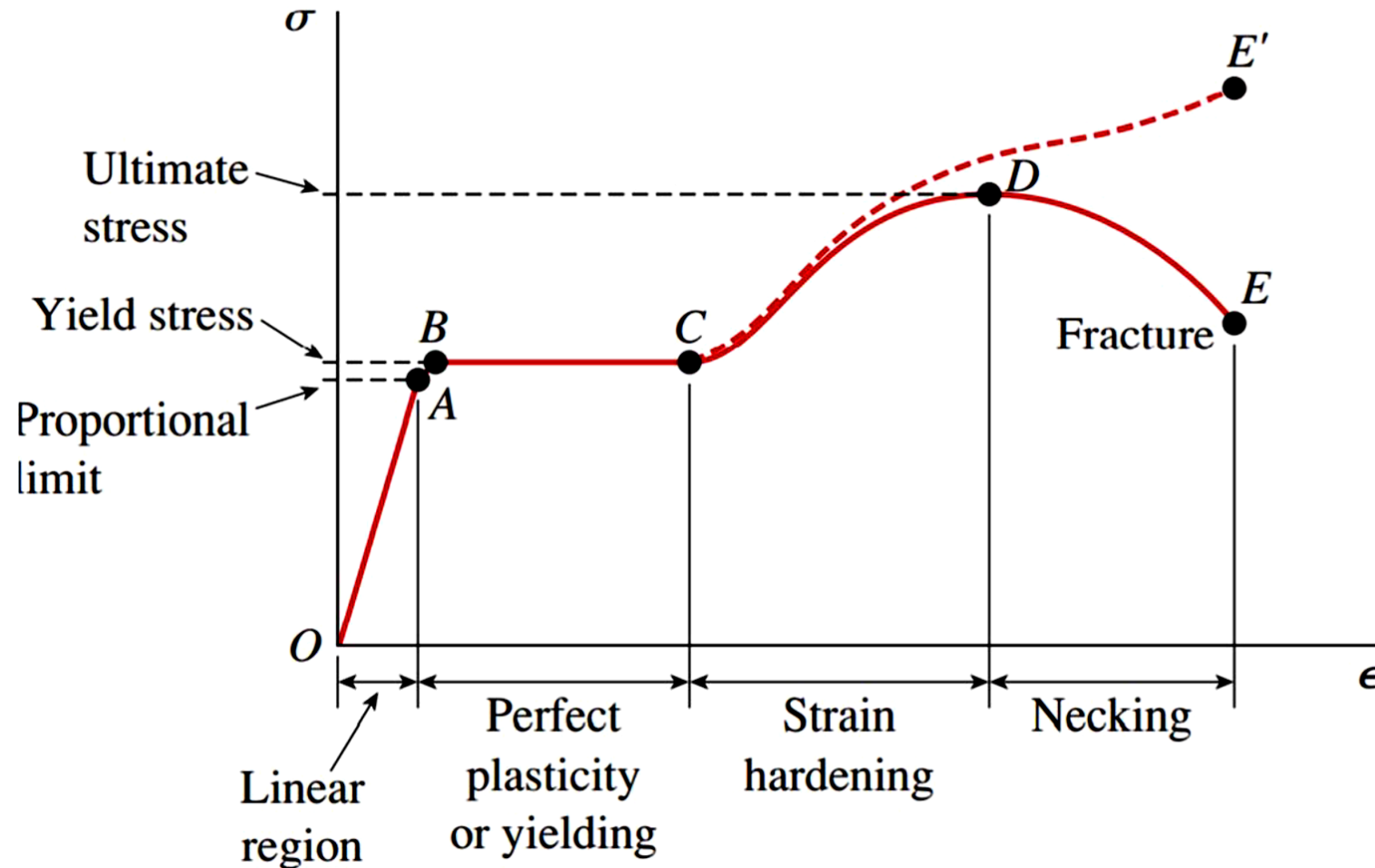
Strain Hardening

During **strain hardening**, the material undergoes changes in its crystalline structure, resulting in increased resistance of the material to further deformation. Elongation of the test specimen in this region requires an increase in the tensile load, and therefore the stress-strain diagram has a positive slope from C to D.



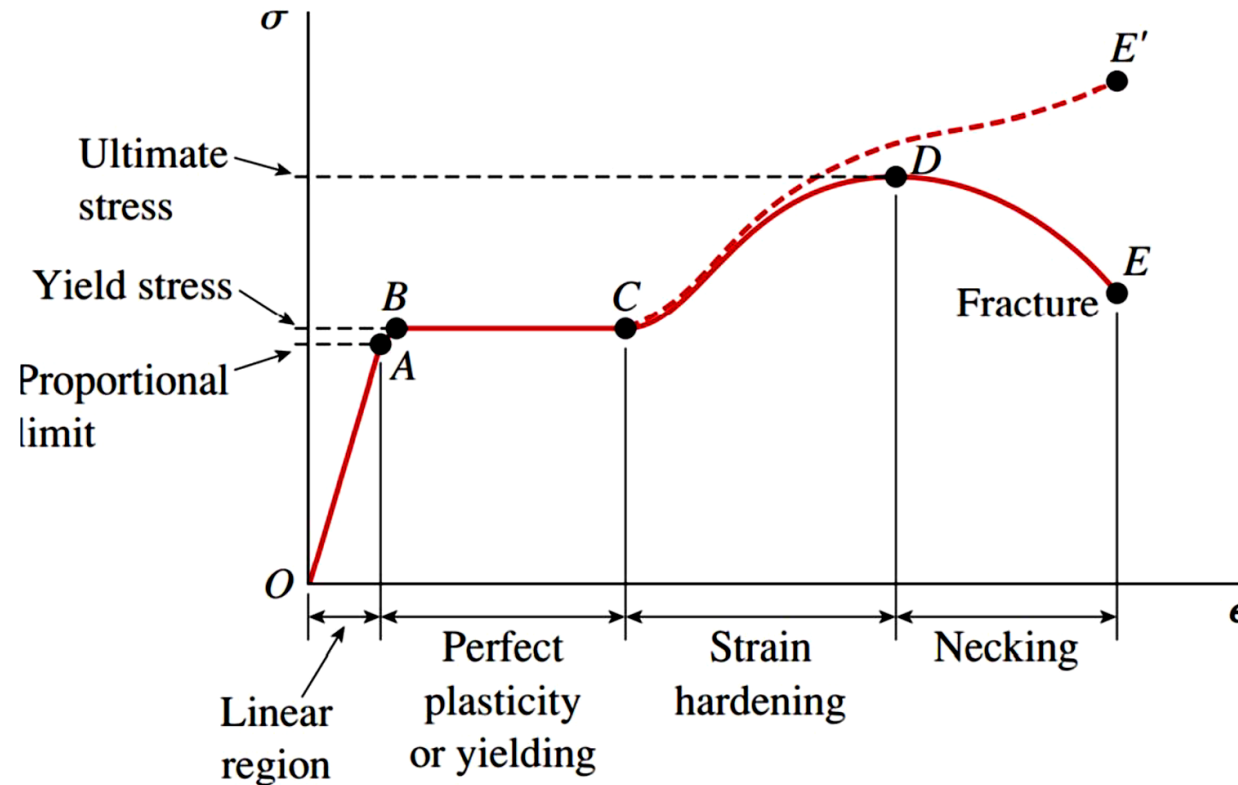
The Yield Stress and Ultimate stress

The yield stress and ultimate stress of a material are also called the yield strength and ultimate strength, respectively.



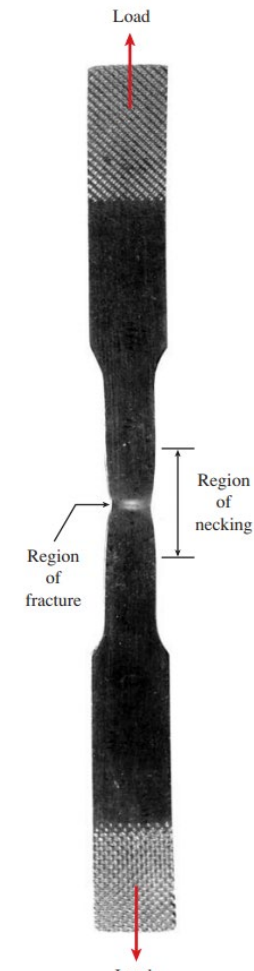
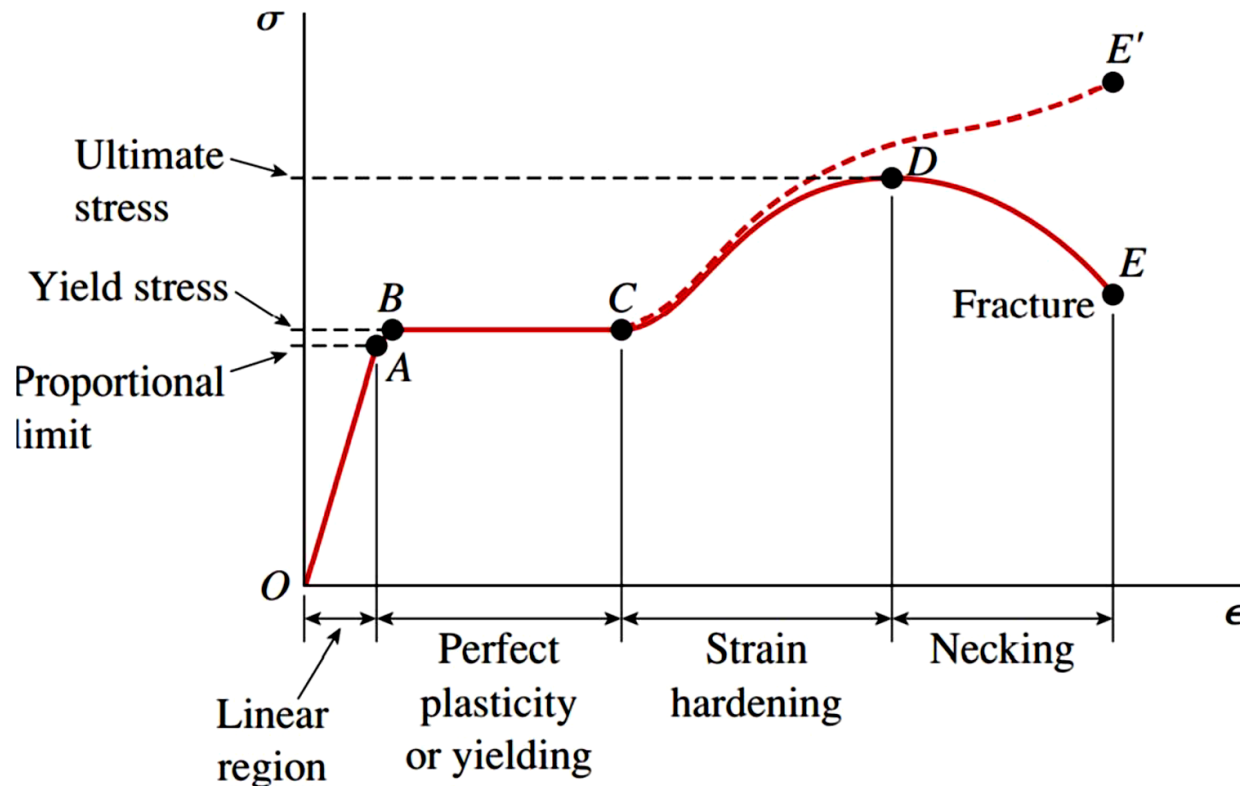
The Meaning of Strength

- Strength is a general term that refers to the capacity of a structure to resist loads.
- For instance, the yield strength of a beam is the magnitude of the load required to cause yielding in the beam, and the ultimate strength of a truss is the maximum load it can support, that is, the failure load.



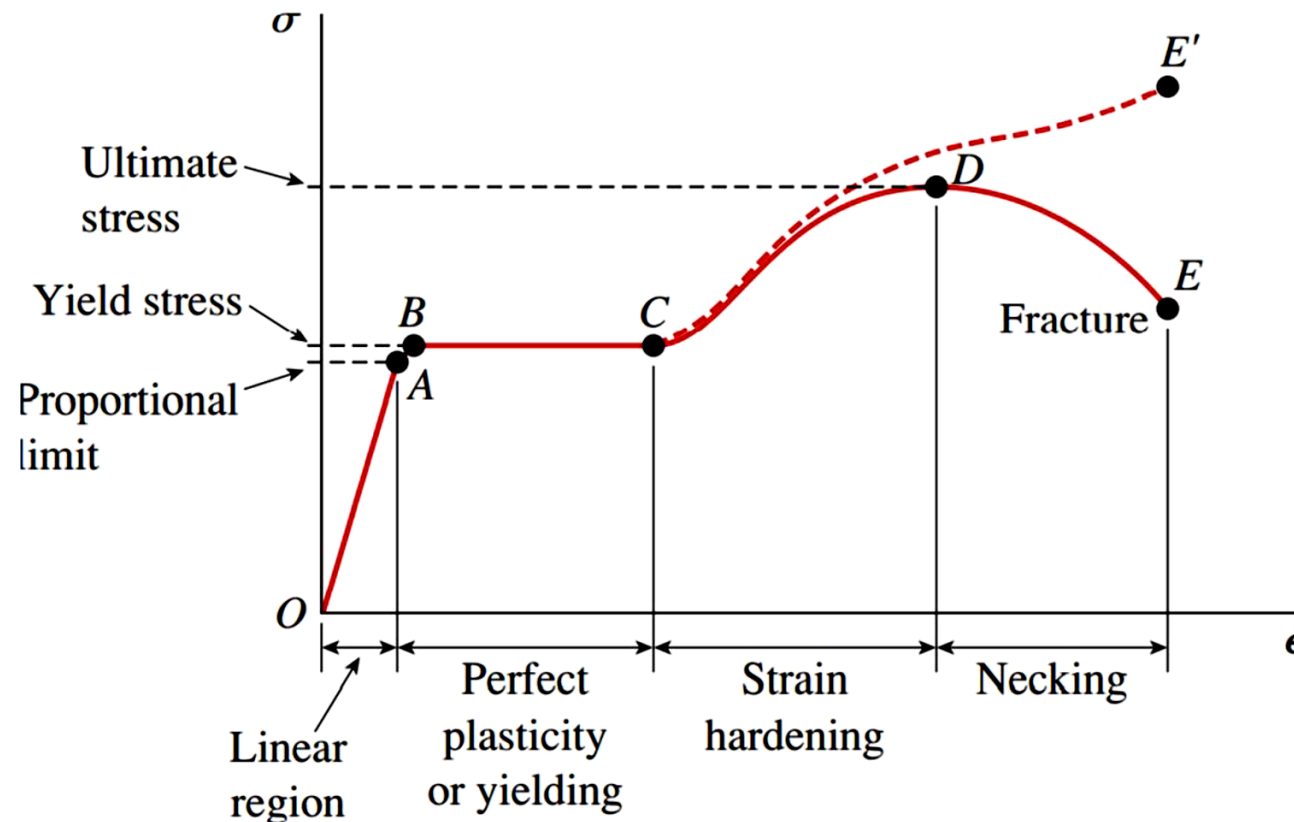
Lateral Contraction

- When a test specimen is stretched, lateral contraction occurs, as
- previously mentioned. The resulting decrease in cross-sectional area is too small to have a noticeable effect on the calculated values of the stresses up to about point C

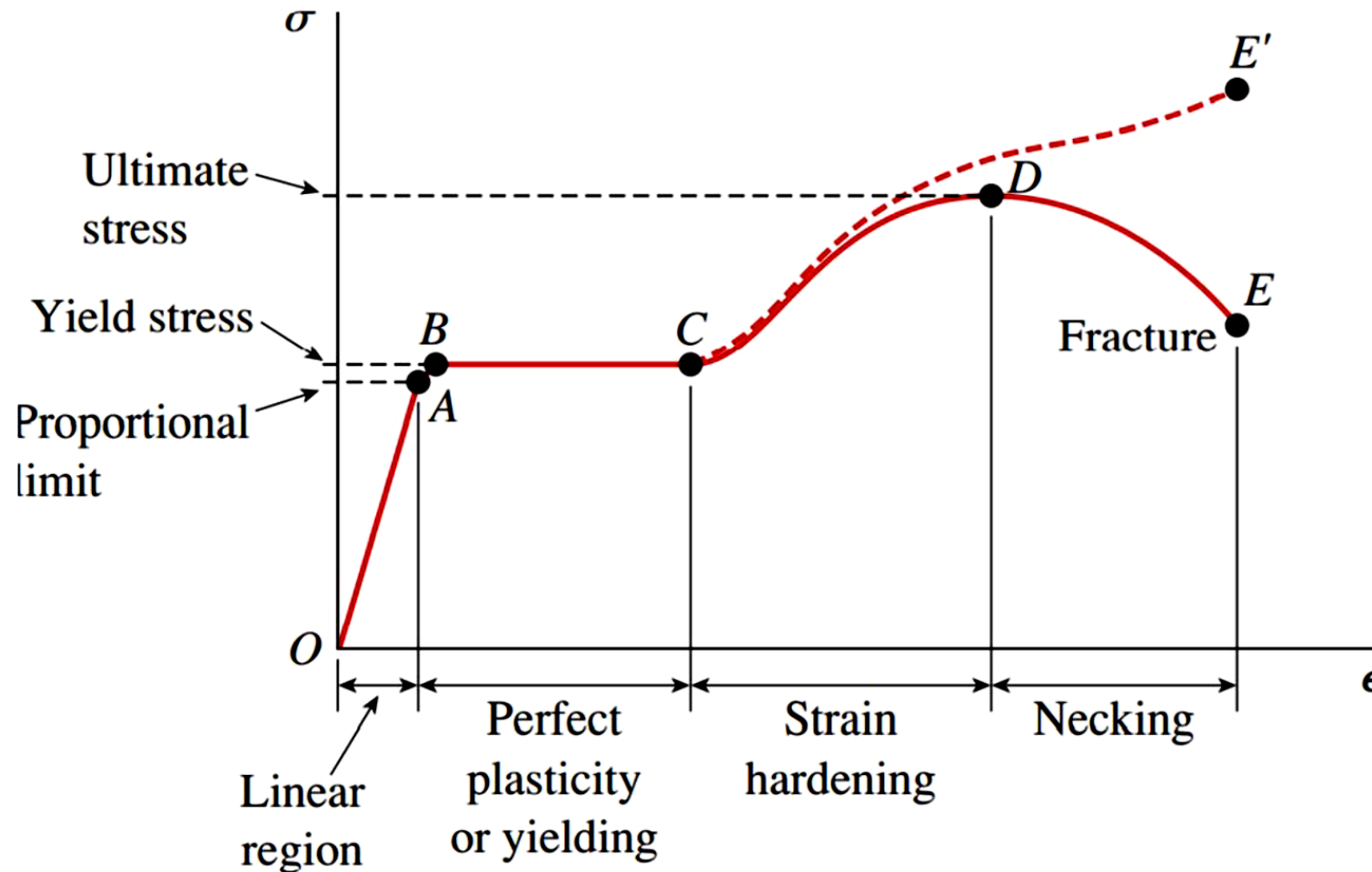


The actual cross-sectional area

If the actual cross-sectional area at the narrow part of the neck is used to calculate the stress, the true stress-strain curve (the dashed line CE) is obtained. The total load the bar can carry does indeed diminish after the ultimate stress is reached (as shown by curve DE), but this reduction is due to the decrease in area of the bar and not to a loss in strength of the material itself



In reality, the material withstands an increase in true stress up to failure (point E).



Different Stress-Strain Curves

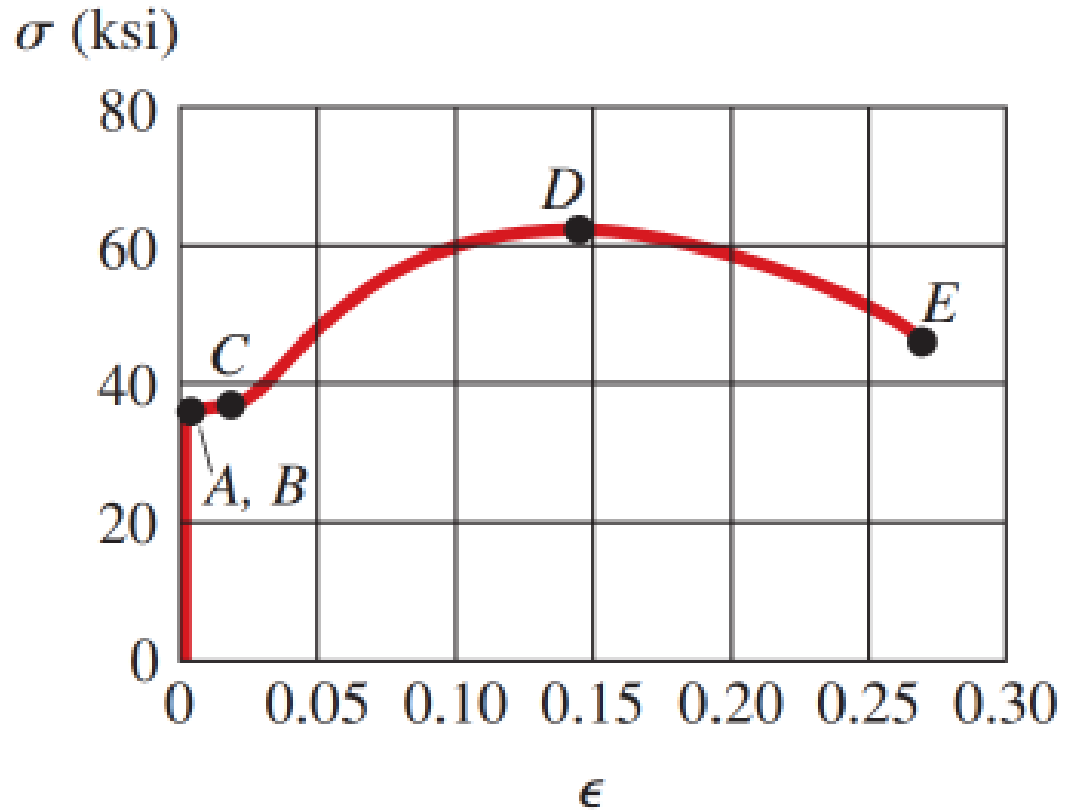


FIG. 1-12 Stress-strain diagram for a typical structural steel in tension (drawn to scale)

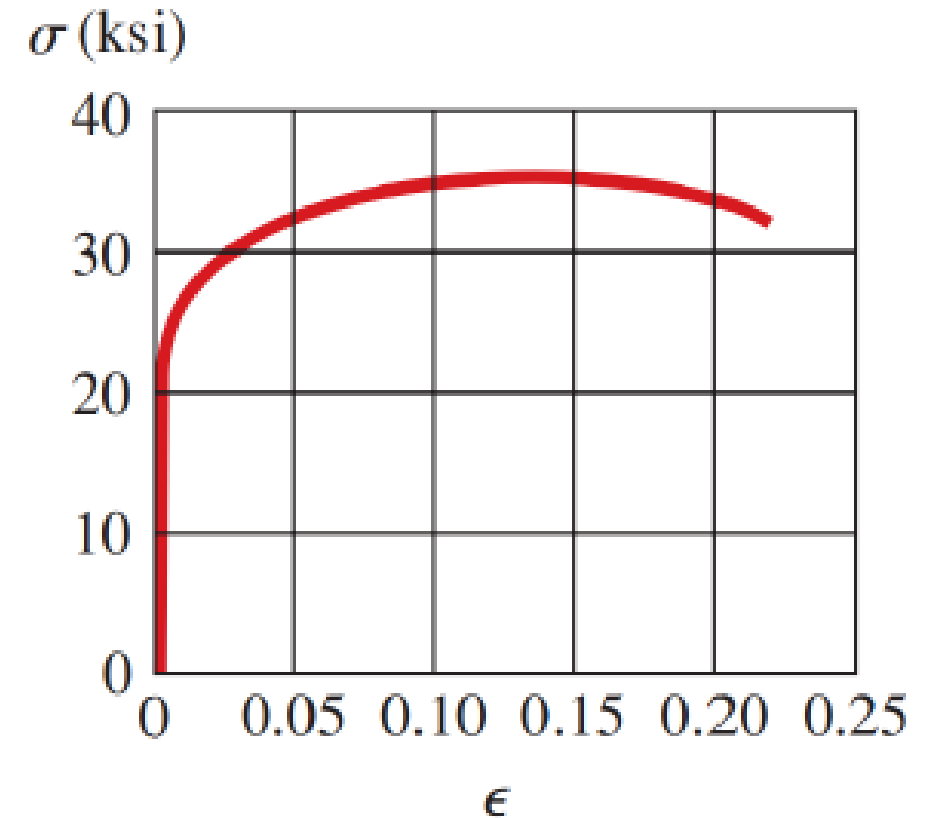


FIG. 1-13 Typical stress-strain diagram for an aluminum alloy

Ductility

The ductility of a material in tension can be characterized by its elongation and by the reduction in area at the cross section where fracture occurs. The **percent elongation** is defined as follows:

$$\text{Percent elongation} = \frac{L_1 - L_0}{L_0} (100) \quad (1-6)$$

in which L_0 is the original gage length and L_1 is the distance between the gage marks at fracture

The **percent reduction in area** measures the amount of necking that occurs and is defined as follows:

$$\text{Percent reduction in area} = \frac{A_0 - A_1}{A_0} (100) \quad (1-7)$$

in which A_0 is the original cross-sectional area and A_1 is the final area at the fracture section. For ductile steels, the reduction is about 50%.

Brittle Materials

- Materials that fail in tension at relatively low values of strain are classified as brittle.
- Examples are concrete, stone, cast iron, glass, ceramics, and a variety of metallic alloys.
- Brittle materials fail with only little elongation after the proportional limit (the stress at point A) is exceeded. Furthermore, the reduction in area is insignificant, and so the nominal fracture stress (point B) is the same as the true ultimate stress.

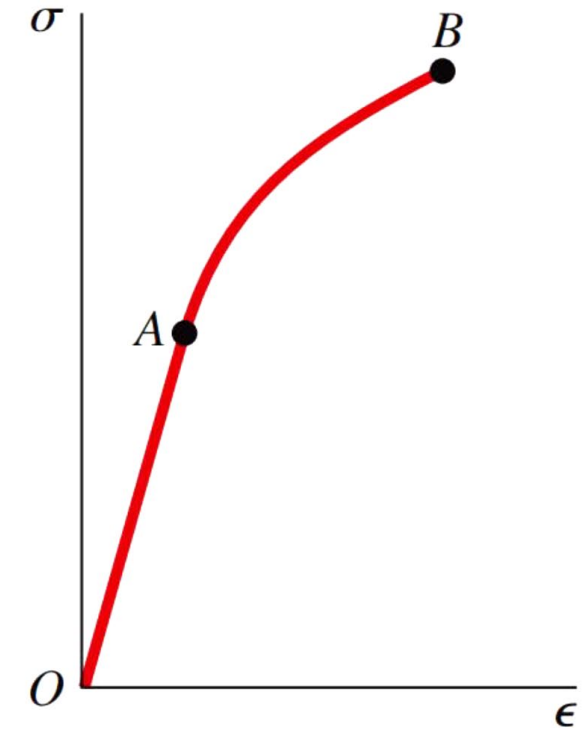


FIG. 1-16 Typical stress-strain diagram for a brittle material showing the proportional limit (point A) and fracture stress (point B)

Compression

- Stress-strain curves for materials in compression differ from those in tension. Ductile metals such as steel, aluminum, and copper have proportional limits in compression very close to those in tension, and the initial regions of their compressive and tensile stress-strain diagrams are about the same. However, after yielding begins, the behavior is quite different.
- In a tension test, the specimen is stretched, necking may occur, and fracture ultimately takes place. When the material is compressed, it bulges outward on the sides and becomes barrel shaped, because friction between the specimen and the end plates prevents lateral expansion.

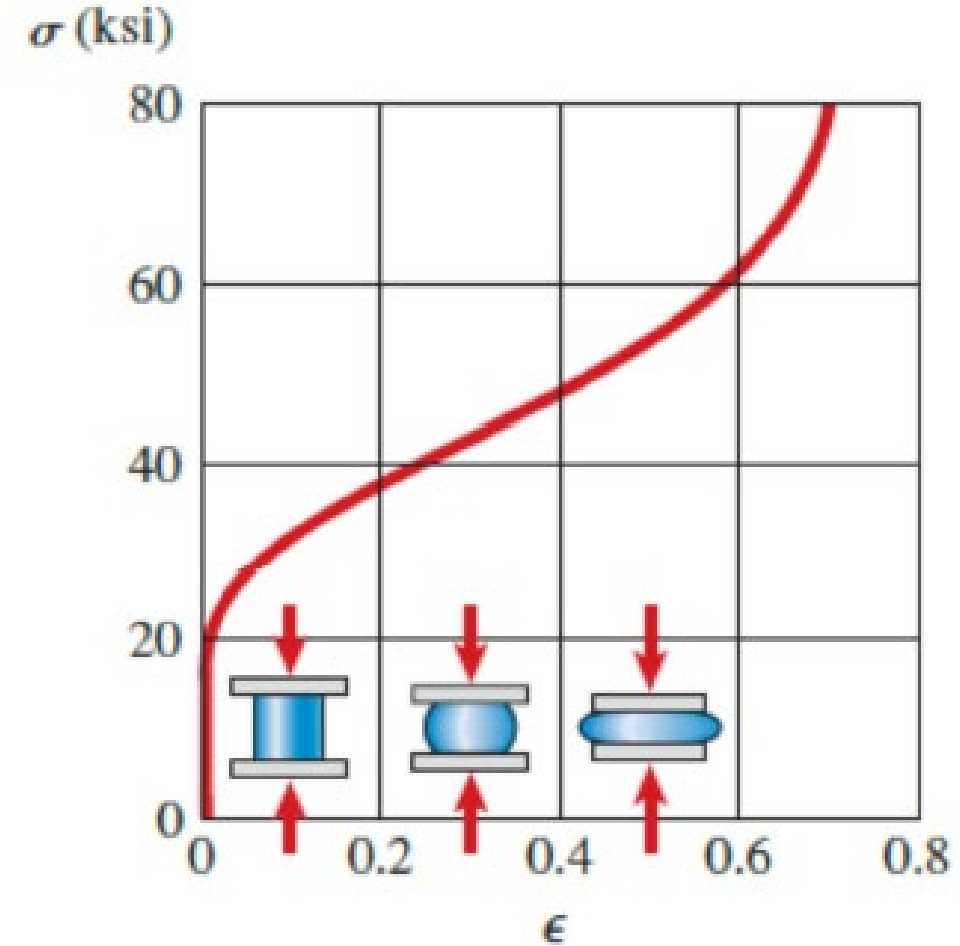


FIG. 1-17 Stress-strain diagram for copper in compression

Hook's Law : Modulus of Elasticity

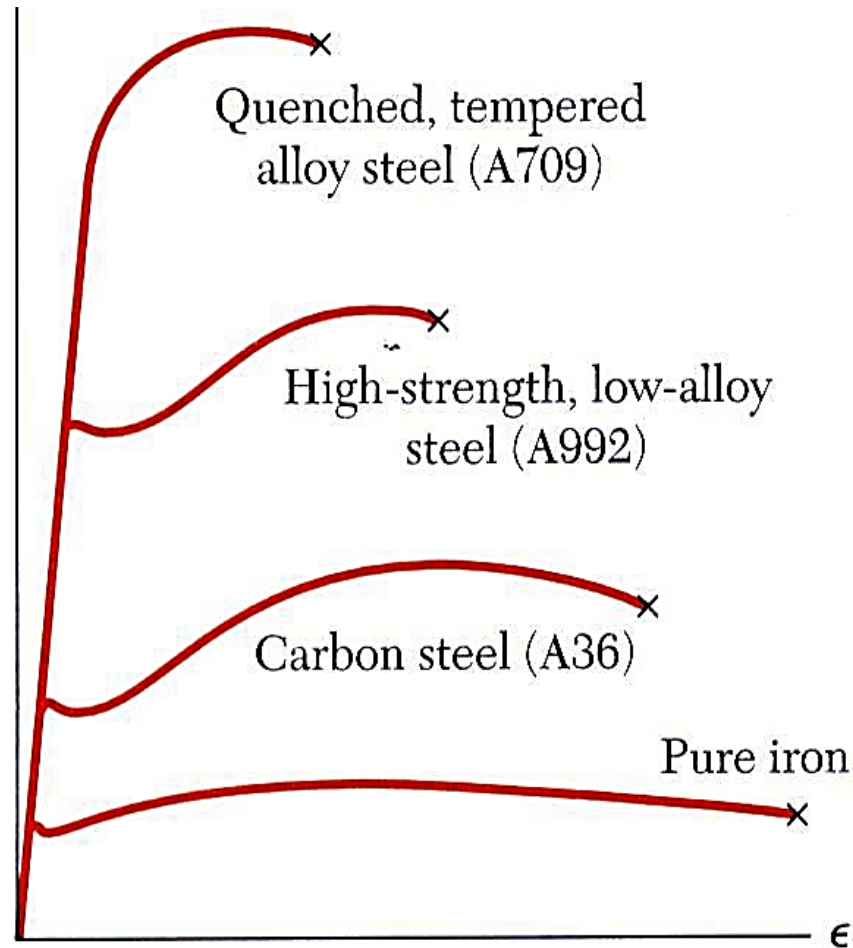


Fig. 2.16 Stress-strain diagrams for iron and different grades of steel.

- Below the yield stress

$$\sigma = E \epsilon$$

E = Youngs Modulus or Modulus of Elasticity

- Strength is affected by alloying, heat treating, and manufacturing process but stiffness (Modulus of Elasticity) is not.

Elastic vs. Plastic Behavior

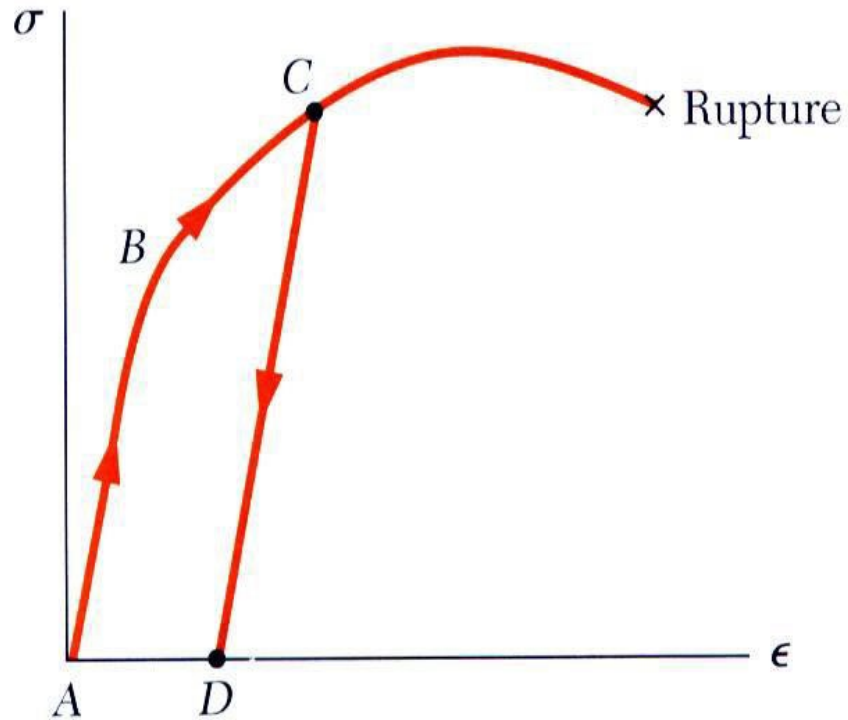


Fig. 2.18

- If the strain disappears when the stress is removed, the material is said to behave *elastically*.
- The largest stress for which this occurs is called the *elastic limit*.
- When the strain does not return to zero after the stress is removed, the material is said to behave *plastically*.

Fatigue

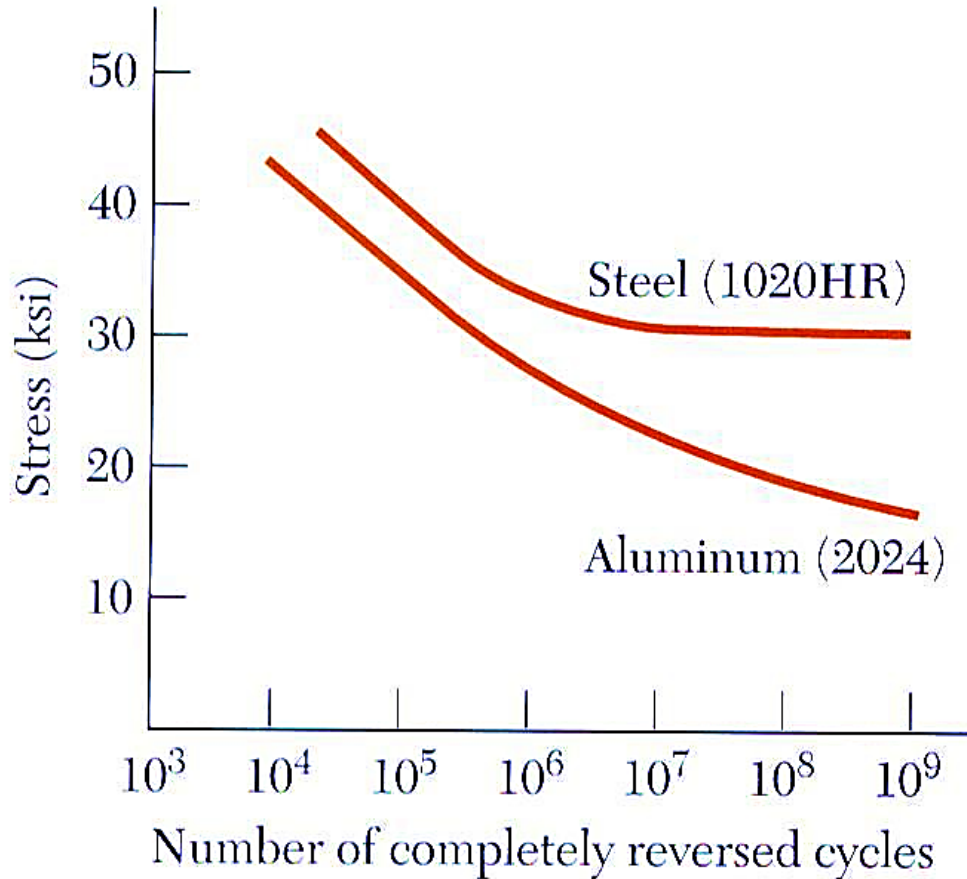


Fig. 2.21

- Fatigue properties are shown on S-N diagrams.
- A member may fail due to *fatigue* at stress levels significantly below the ultimate strength if subjected to many loading cycles.
- When the stress is reduced below the *endurance limit*, fatigue failures do not occur for any number of cycles.

Deformation on Axial Loading

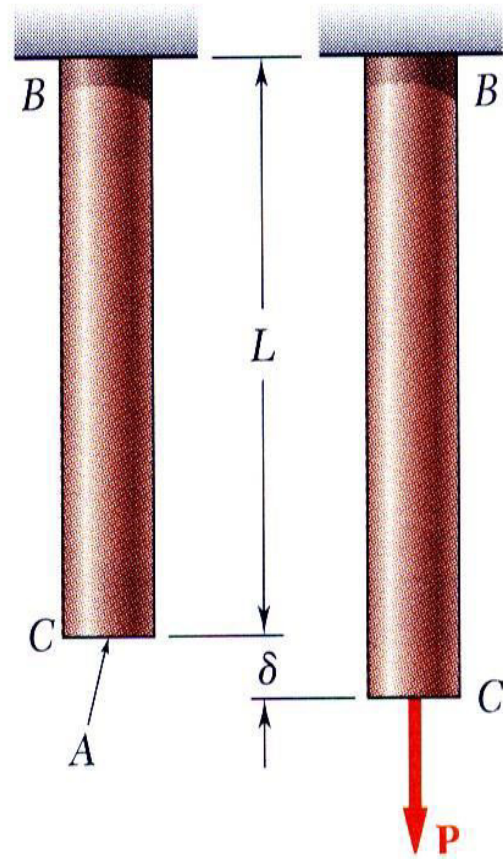


Fig. 2.22

- From Hooke's Law:

$$\sigma = E\varepsilon \quad \varepsilon = \frac{\sigma}{E} = \frac{P}{AE}$$

- From the definition of strain:

$$\varepsilon = \frac{\delta}{L}$$

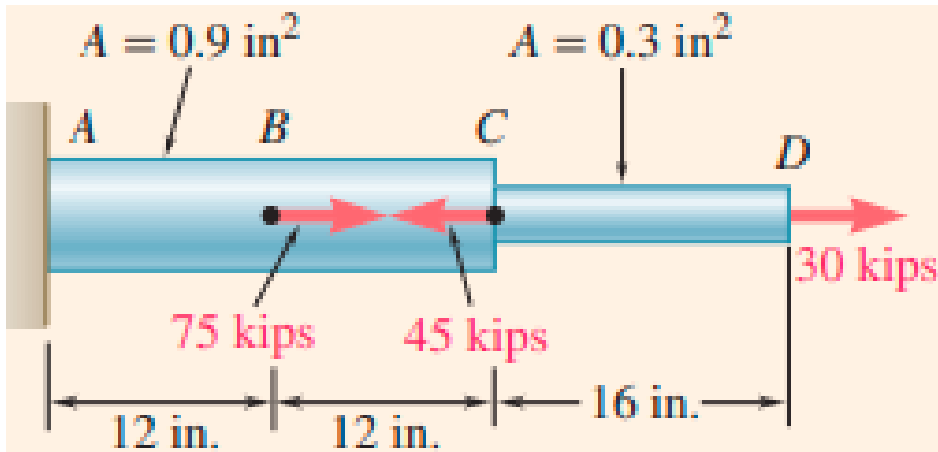
- Equating and solving for the deformation,

$$\delta = \frac{PL}{AE}$$

- With variations in loading, cross-section or material properties,

$$\delta = \sum_i \frac{P_i L_i}{A_i E_i}$$

Example



$$E = 29 \times 10^6 \text{ psi}$$

$$D = 1.07 \text{ in. } d = 0.618 \text{ in.}$$

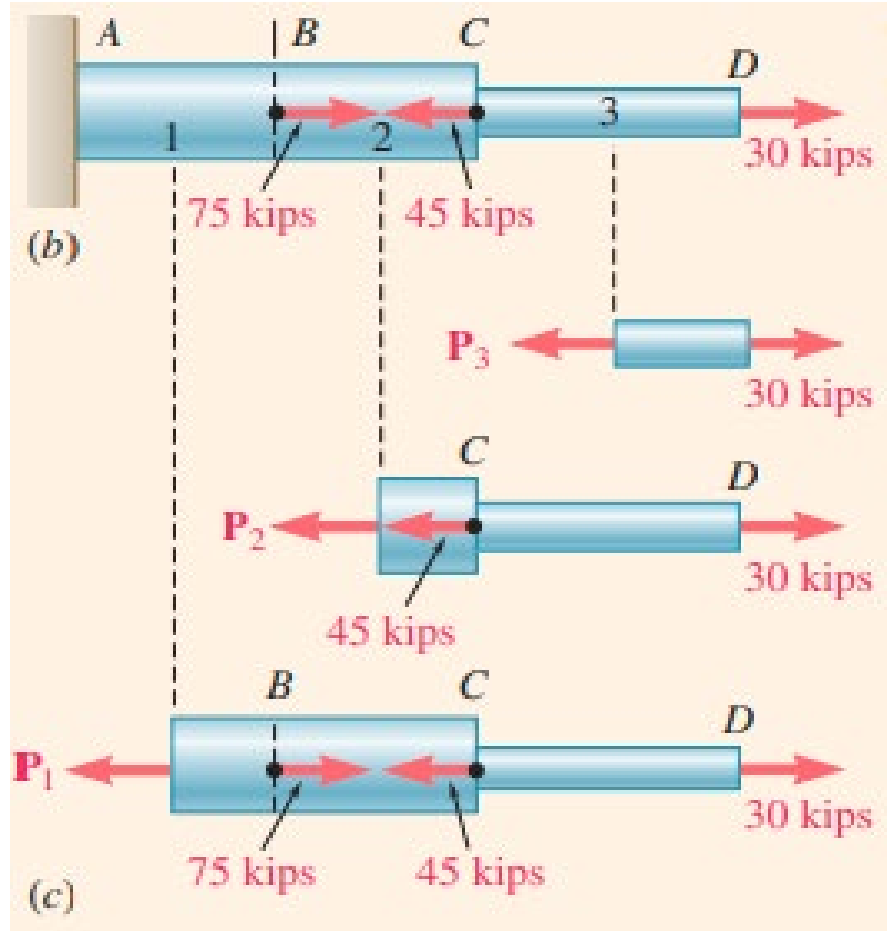
Solution:

- Divide the load into components at the load application points.
- Apply a free-body analysis on each component to determine the internal force.
- Evaluate the total of the components deflections

Determine the deformation of the steel rod shown under the given loads.

SOLUTION:

- Divide the rod into three components:



$$L_1 = L_2 = 12 \text{ in.} \quad L_3 = 16 \text{ in.}$$

$$A_1 = A_2 = 0.9 \text{ in}^2 \quad A_3 = 0.3 \text{ in}^2$$

- Apply free-body analysis to each component to determine internal forces,

$$P_1 = 60 \times 10^3 \text{ lb}$$

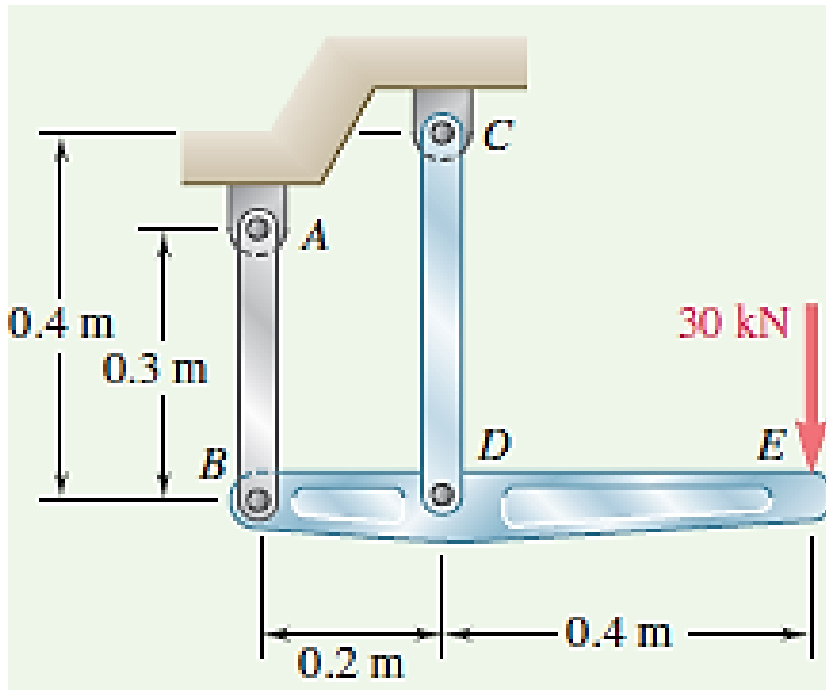
$$P_2 = -15 \times 10^3 \text{ lb}$$

$$P_3 = 30 \times 10^3 \text{ lb}$$

- Evaluate total deflection,

$$\begin{aligned} \delta &= \sum_i \frac{P_i L_i}{A_i E_i} = \frac{1}{E} \left(\frac{P_1 L_1}{A_1} + \frac{P_2 L_2}{A_2} + \frac{P_3 L_3}{A_3} \right) \\ &= \frac{1}{29 \times 10^6} \left[\frac{(60 \times 10^3) 12}{0.9} + \frac{(-15 \times 10^3) 12}{0.9} + \frac{(30 \times 10^3) 16}{0.3} \right] \\ &= 75.9 \times 10^{-3} \text{ in.} \end{aligned}$$

Example



The rigid bar BDE is supported by two links AB and CD .

Link AB is made of aluminum ($E = 70\text{ GPa}$) and has a cross-sectional area of 500 mm^2 . Link CD is made of steel ($E = 200\text{ GPa}$) and has a cross-sectional area of (600 mm^2).

For the 30-kN force shown, determine the deflection a) of B , b) of D , and c) of E .

SOLUTION:

- Apply a free-body analysis to the bar BDE to find the forces exerted by links AB and DC .
- Evaluate the deformation of links AB and DC or the displacements of B and D .
- Work out the geometry to find the deflection at E given the deflections at B and D .

SOLUTION:

Free body: Bar *BDE*

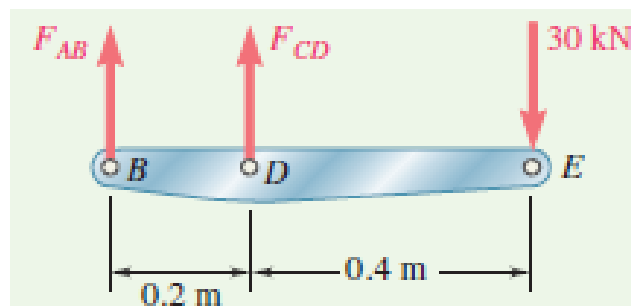


Fig. 1 Free-body diagram of rigid bar *BDE*.

$$\sum M_B = 0$$

$$0 = -(30 \text{ kN} \times 0.6 \text{ m}) + F_{CD} \times 0.2 \text{ m}$$

$$F_{CD} = +90 \text{ kN} \text{ tension}$$

$$\sum M_D = 0$$

$$0 = -(30 \text{ kN} \times 0.4 \text{ m}) - F_{AB} \times 0.2 \text{ m}$$

$$F_{AB} = -60 \text{ kN} \text{ compression}$$

Displacement of *B*:

$$\delta_B = \frac{PL}{AE}$$

$$= \frac{(-60 \times 10^3 \text{ N})(0.3 \text{ m})}{(500 \times 10^{-6} \text{ m}^2)(70 \times 10^9 \text{ Pa})}$$

$$= -514 \times 10^{-6} \text{ m}$$

$\delta_B = 0.514 \text{ mm} \uparrow$

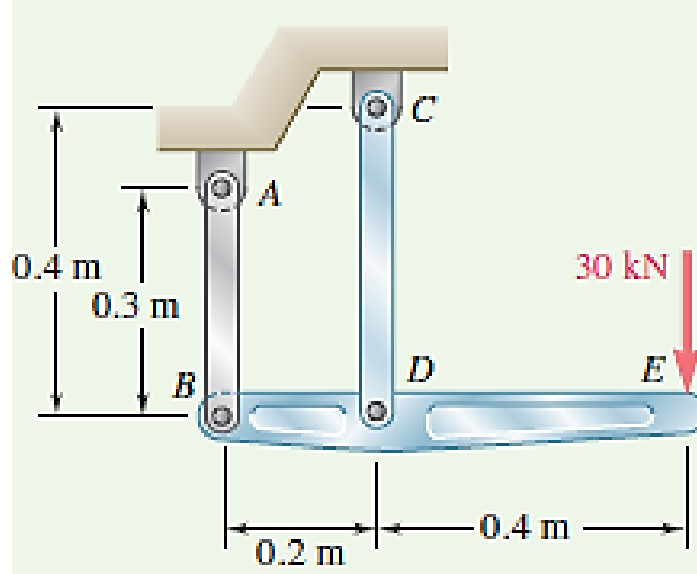
Displacement of *D*:

$$\delta_D = \frac{PL}{AE}$$

$$= \frac{(90 \times 10^3 \text{ N})(0.4 \text{ m})}{(600 \times 10^{-6} \text{ m}^2)(200 \times 10^9 \text{ Pa})}$$

$$= 300 \times 10^{-6} \text{ m}$$

$\delta_D = 0.300 \text{ mm} \downarrow$



Displacement of D:

$$\frac{BB'}{DD'} = \frac{BH}{HD}$$

$$\frac{0.514 \text{ mm}}{0.300 \text{ mm}} = \frac{(200 \text{ mm}) - x}{x}$$

$$x = 73.7 \text{ mm}$$

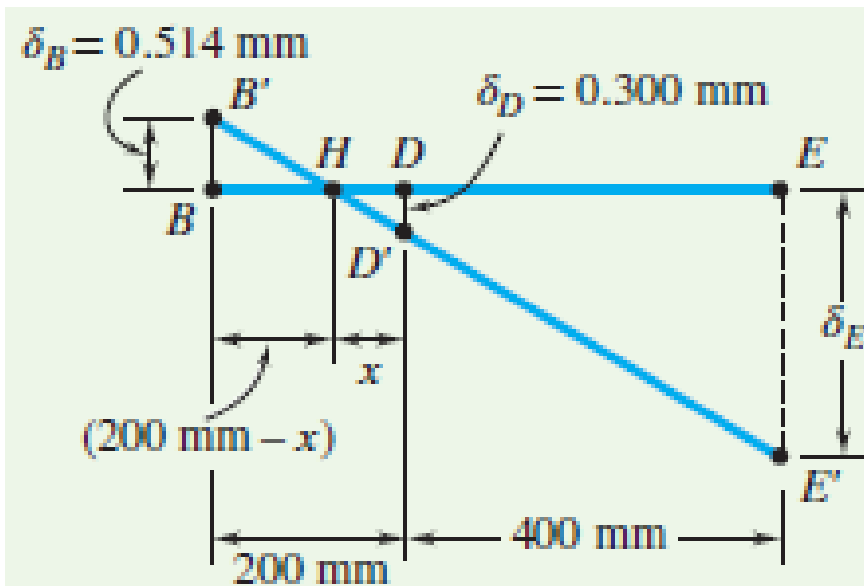


Fig. 4 Deflections at *B* and *D* of rigid bar are used to find δ_E .

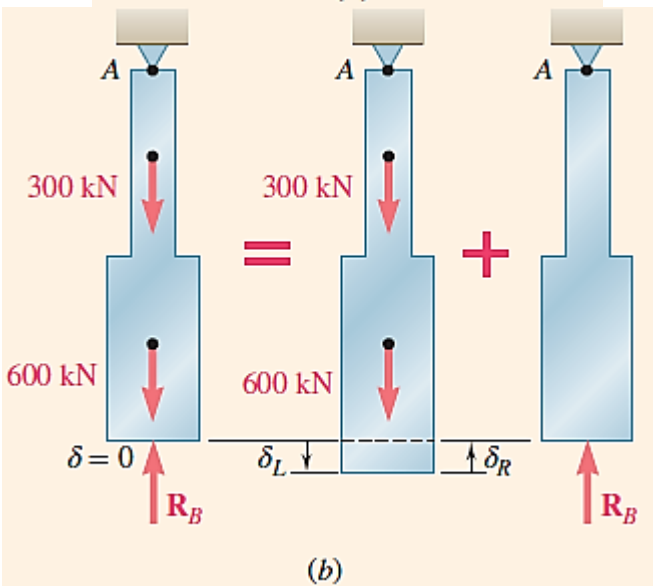
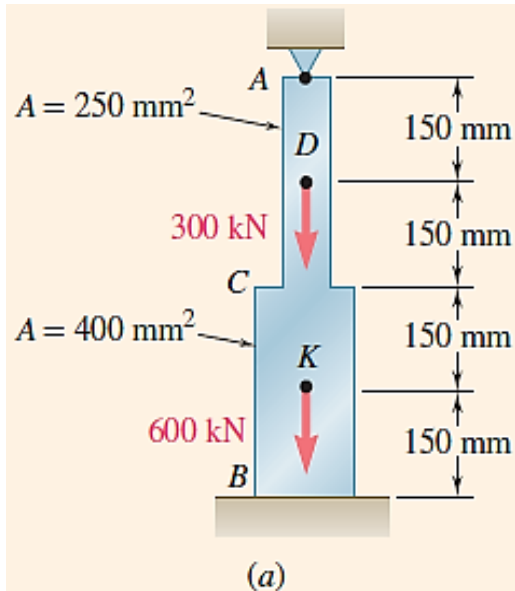
$$\frac{EE'}{DD'} = \frac{HE}{HD}$$

$$\frac{\delta_E}{0.300 \text{ mm}} = \frac{(400 + 73.7) \text{ mm}}{73.7 \text{ mm}}$$

$$\delta_E = 1.928 \text{ mm}$$

$$\boxed{\delta_E = 1.928 \text{ mm} \downarrow}$$

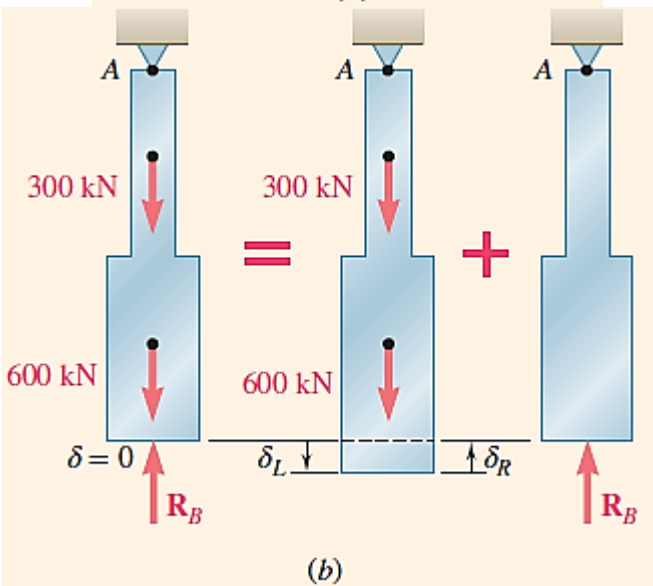
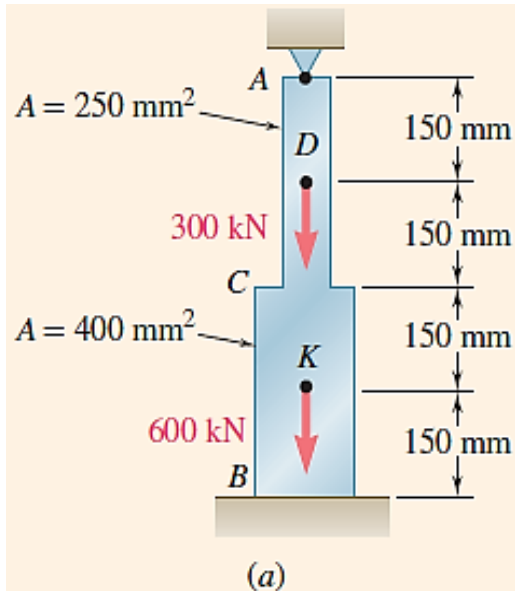
Static Indeterminacy



- Structures for which internal forces and reactions cannot be determined from statics alone are said to be *statically indeterminate*.
- A structure will be statically indeterminate whenever it is held by more supports than are required to maintain its equilibrium.
- Redundant reactions** are replaced with unknown loads which along with the other loads must produce compatible deformations.
- Deformations due to actual loads and redundant reactions are determined separately and then added or *superposed*.

$$\delta = \delta_L + \delta_R = 0$$

Example



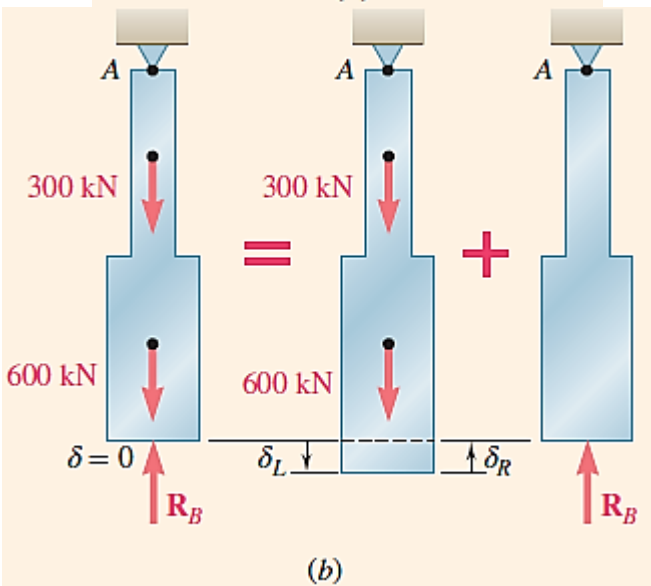
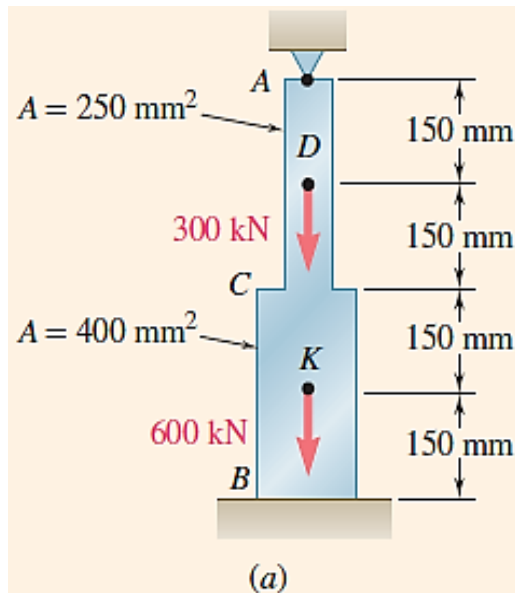
Determine the reactions at A and B for the steel bar and loading shown, assuming a close fit at both supports before the loads are applied.

SOLUTION:

- Consider the reaction at B as redundant, release the bar from that support, and solve for the displacement at B due to the applied loads.
- Solve for the displacement at B due to the redundant reaction at B.
- Require that the displacements due to the loads and due to the redundant reaction be compatible, i.e., require that their sum be zero.
- Solve for the reaction at A due to applied loads and the reaction found at B.

Example

SOLUTION:



- Solve for the displacement at B due to the applied loads with the redundant constraint released,

$$P_1 = 0 \quad P_2 = P_3 = 600 \times 10^3 \text{ N} \quad P_4 = 900 \times 10^3 \text{ N}$$

$$A_1 = A_2 = 400 \times 10^{-6} \text{ m}^2 \quad A_3 = A_4 = 250 \times 10^{-6} \text{ m}^2$$

$$L_1 = L_2 = L_3 = L_4 = 0.150 \text{ m}$$

$$\delta_L = \sum_i \frac{P_i L_i}{A_i E_i} = \frac{1.125 \times 10^9}{E}$$

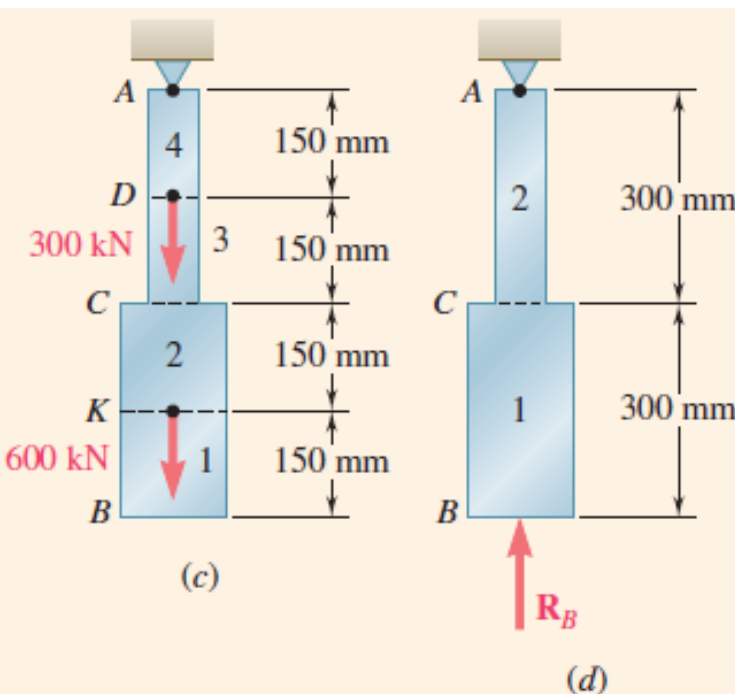
- Solve for the displacement at B due to the redundant constraint,

$$P_1 = P_2 = -R_B$$

$$A_1 = 400 \times 10^{-6} \text{ m}^2 \quad A_2 = 250 \times 10^{-6} \text{ m}^2$$

$$L_1 = L_2 = 0.300 \text{ m}$$

$$\delta_R = \sum_i \frac{P_i L_i}{A_i E_i} = -\frac{(1.95 \times 10^3) R_B}{E}$$



- Require that the displacements due to the loads and due to the redundant reaction be compatible,

$$\delta = \delta_L + \delta_R = 0$$

$$\delta = \frac{1.125 \times 10^9}{E} - \frac{(1.95 \times 10^3) R_B}{E} = 0$$

$$R_B = 577 \times 10^3 \text{ N} = 577 \text{ kN}$$

- Find the reaction at A due to the loads and the reaction at B

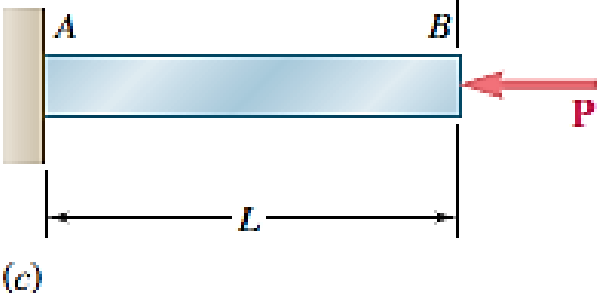
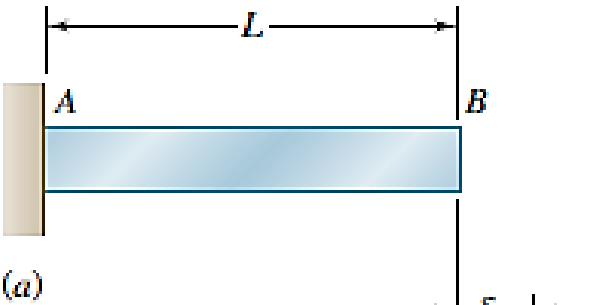
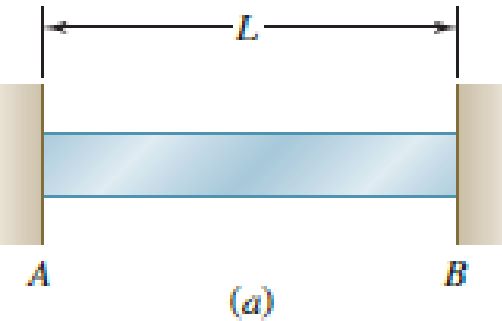
$$\sum F_y = 0 = R_A - 300 \text{ kN} - 600 \text{ kN} + 577 \text{ kN}$$

$$R_A = 323 \text{ kN}$$

$$R_A = 323 \text{ kN}$$

$$R_B = 577 \text{ kN}$$

Thermal Stresses



- A temperature change results in a change in length or *thermal strain*. There is no stress associated with the thermal strain unless the elongation is restrained by the supports.

- Treat the additional support as redundant and apply the principle of superposition.

$$\delta_T = \alpha(\Delta T)L$$

$$\delta_P = \frac{PL}{AE}$$

α = thermal expansion coef.

- The thermal deformation and the deformation from the redundant support must be compatible.

$$\delta = \delta_T + \delta_P = 0$$

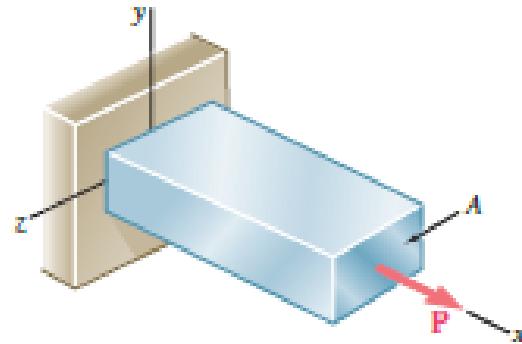
$$\alpha(\Delta T)L + \frac{PL}{AE} = 0$$

$$\delta = \delta_T + \delta_P = 0$$

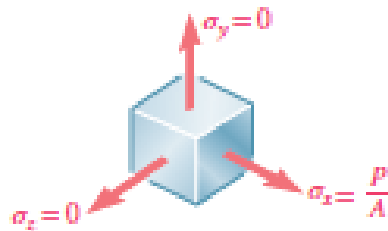
$$P = -AE\alpha(\Delta T)$$

$$\sigma = \frac{P}{A} = -E\alpha(\Delta T)$$

Poisson's Ratio



(a)



(b)

Fig. 2.29 A bar in uniaxial tension and a representative stress element.

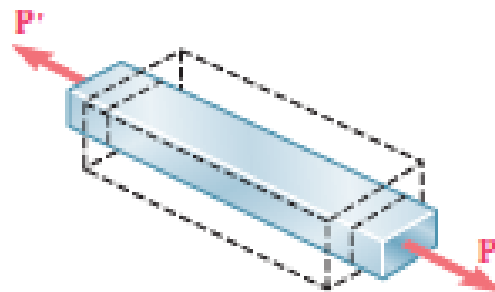


Fig. 2.30 Materials undergo transverse contraction when elongated under axial load.

- For a slender bar subjected to axial loading:

$$\epsilon_x = \frac{\sigma_x}{E} \quad \sigma_y = \sigma_z = 0$$

- The elongation in the x-direction is accompanied by a contraction in the other directions. Assuming that the material is isotropic (no directional dependence),

$$\epsilon_y = \epsilon_z \neq 0$$

- Poisson's ratio is defined as

$$\nu = \left| \frac{\text{lateral strain}}{\text{axial strain}} \right| = -\frac{\epsilon_y}{\epsilon_x} = -\frac{\epsilon_z}{\epsilon_x}$$

Poisson's Ratio

- For isotropic materials, Poisson found $\nu = 1/4$
- More recent calculations based upon better models of atomic structure give $\nu=1/3$
- Both of these values are close to actual measured values, which are in the range 0.25 to 0.35 for most metals and many other materials.
- Materials with an extremely low value of Poisson's ratio include cork, for which ν is practically zero, and concrete, for which ν is about 0.1 or 0.2.
- A theoretical upper limit for Poisson's ratio is 0.5, as explained later. Rubber comes close to this limiting value
- For most purposes, Poisson's ratio is assumed to be the same in both tension and compression.
- When the **strains** in a material become large, Poisson's ratio changes. For instance, in the case of structural steel, the ratio becomes almost 0.5 when plastic yielding occurs. Thus, Poisson's ratio remains constant only in the linearly elastic range. When the material behavior is nonlinear, the ratio of lateral strain to axial strain is often called the contraction ratio. Of course, in the special case of linearly elastic behavior, the contraction ratio is the same as Poisson's ratio.

Generalized Hooke's Law

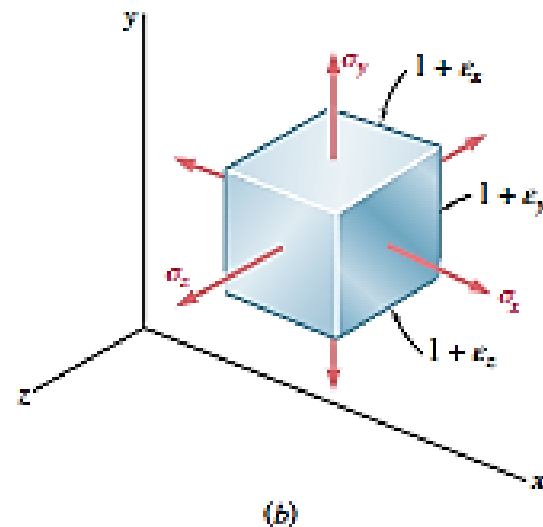
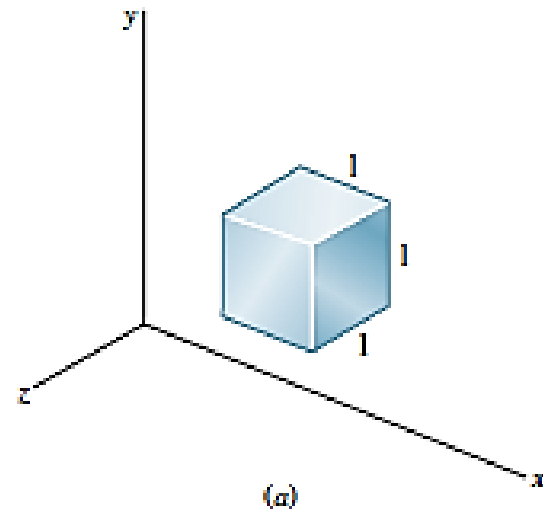


Fig. 2.33 Deformation of unit cube under multiaxial loading: (a) unloaded; (b) deformed.

- For an element subjected to multi-axial loading, the normal strain components resulting from the stress components may be determined from the *principle of superposition*. This requires:

- 1) strain is linearly related to stress
- 2) deformations are small

- With these restrictions:

$$\varepsilon_x = +\frac{\sigma_x}{E} - \frac{\nu\sigma_y}{E} - \frac{\nu\sigma_z}{E}$$

$$\varepsilon_y = -\frac{\nu\sigma_x}{E} + \frac{\sigma_y}{E} - \frac{\nu\sigma_z}{E}$$

$$\varepsilon_z = -\frac{\nu\sigma_x}{E} - \frac{\nu\sigma_y}{E} + \frac{\sigma_z}{E}$$

Dilatation: Bulk Modulus

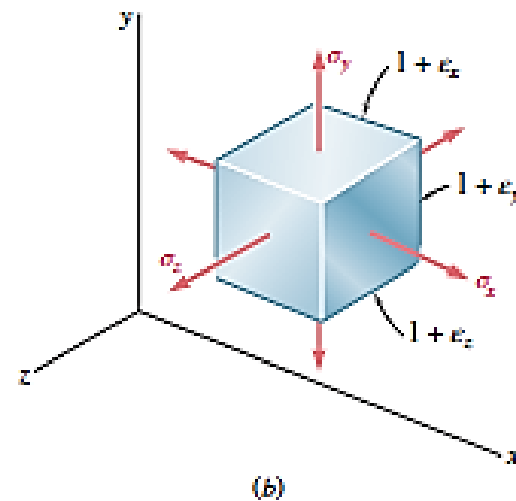
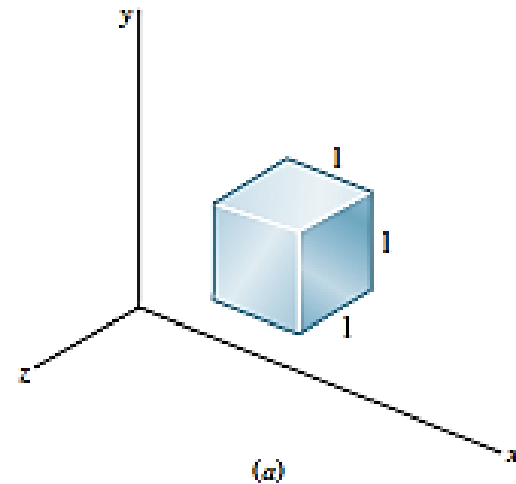


Fig. 2.33 Deformation of unit cube under multiaxial loading: (a) unloaded; (b) deformed.

- Relative to the unstressed state, the change in volume is

$$e = [(1 + \epsilon_x)(1 + \epsilon_y)(1 + \epsilon_z)] - 1 = [1 + \epsilon_x + \epsilon_y + \epsilon_z] - 1$$

$$= \epsilon_x + \epsilon_y + \epsilon_z$$

$$= \frac{1 - 2\nu}{E} (\sigma_x + \sigma_y + \sigma_z)$$
 = dilatation (change in volume per unit volume)

- For element subjected to uniform hydrostatic pressure,

$$e = -p \frac{3(1 - 2\nu)}{E} = -\frac{p}{k}$$

$$k = \frac{E}{3(1 - 2\nu)} = \text{bulk modulus}$$

- Subjected to uniform pressure, dilatation must be negative, therefore

$$0 < \nu < \frac{1}{2}$$

Shearing Strain

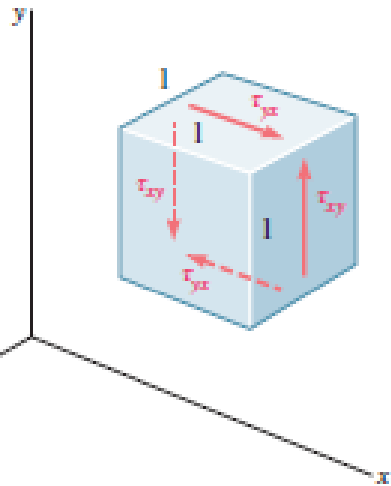


Fig. 2.36 Unit cubic element subjected to shearing stress.

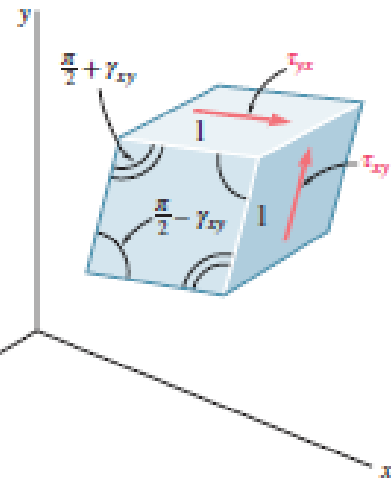


Fig. 2.37 Deformation of unit cubic element due to shearing stress.

- A cubic element subjected to a shear stress will deform into a rhomboid. The corresponding *shear strain* is quantified in terms of the change in angle between the sides,

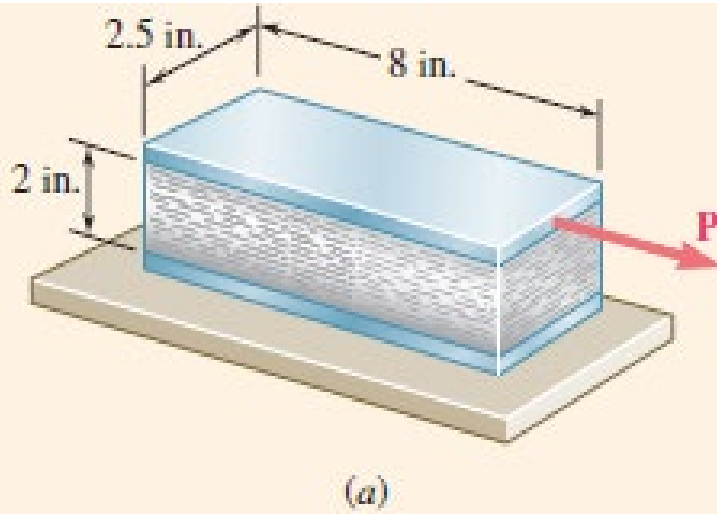
$$\tau_{xy} = f(\gamma_{xy})$$

- A plot of shear stress vs. shear strain is similar the previous plots of normal stress vs. normal strain except that the strength values are approximately half. For small strains,

$$\tau_{xy} = G \gamma_{xy} \quad \tau_{yz} = G \gamma_{yz} \quad \tau_{zx} = G \gamma_{zx}$$

where G is the modulus of rigidity or shear modulus.

Example



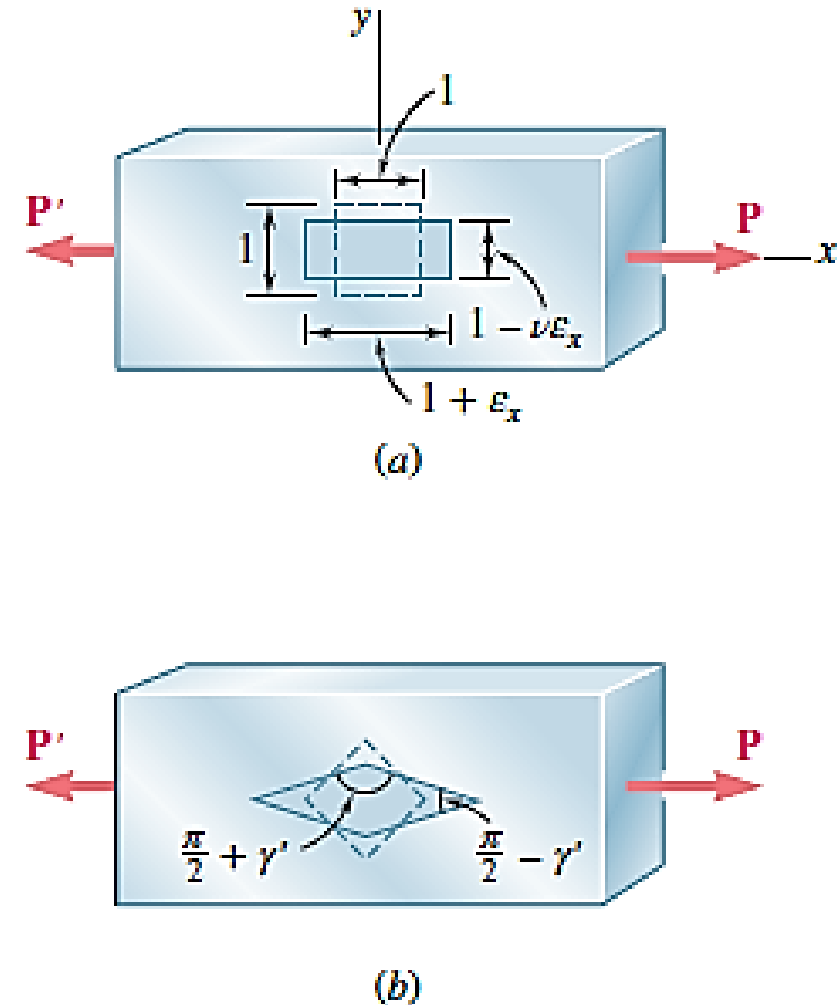
A rectangular block of material with modulus of rigidity $G = 90$ ksi is bonded to two rigid horizontal plates. The lower plate is fixed, while the upper plate is subjected to a horizontal force P . Knowing that the upper plate moves through 0.04 in. under the action of the force, determine

- the average shearing strain in the material, and
- the force P exerted on the plate.

SOLUTION:

- Determine the average angular deformation or shearing strain of the block.
- Apply Hooke's law for shearing stress and strain to find the corresponding shearing stress.
- Use the definition of shearing stress to find the force P .

Relation Among E, ν , and G

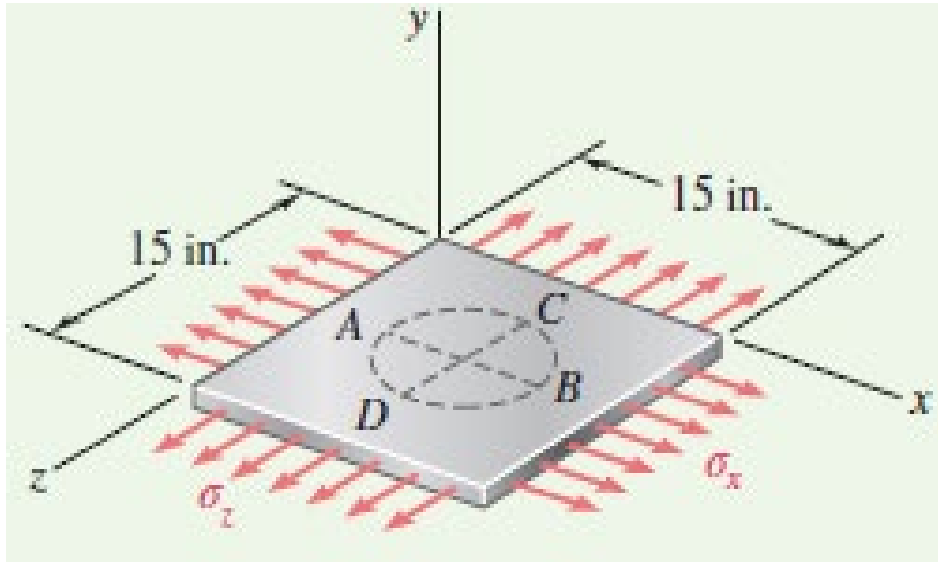


- An axially loaded slender bar will elongate in the axial direction and contract in the transverse directions.
- An initially cubic element oriented as in top figure will deform into a rectangular parallelepiped. The axial load produces a normal strain.
- If the cubic element is oriented as in the bottom figure, it will deform into a rhombus. Axial load also results in a shear strain.
- Components of normal and shear strain are related,

$$\frac{E}{2G} = (1 + \nu)$$

Fig. 2.42 Representations of strain in an axially loaded bar: (a) cubic strain element faces aligned with coordinate axes; (b) cubic strain element faces rotated 45° about z axis.

Example



A circle of diameter $d = 9$ in. is scribed on an unstressed aluminum plate of thickness $t = 3/4$ in. Forces acting in the plane of the plate later cause normal stresses $\sigma_x = 12$ ksi and $\sigma_z = 20$ ksi.

For $E = 10 \times 10^6$ psi and $\nu = 1/3$, determine the change in:

- a) the length of diameter AB ,
- b) the length of diameter CD ,
- c) the thickness of the plate, and
- d) the volume of the plate.

- Apply the generalized Hooke's Law to find the three components of normal strain.
- Evaluate the deformation components.

$$\begin{aligned}\varepsilon_x &= +\frac{\sigma_x}{E} - \frac{\nu\sigma_y}{E} - \frac{\nu\sigma_z}{E} \\ &= \frac{1}{10 \times 10^6 \text{ psi}} \left[(12 \text{ ksi}) - 0 - \frac{1}{3}(20 \text{ ksi}) \right] \\ &= +0.533 \times 10^{-3} \text{ in./in.}\end{aligned}$$

$$\begin{aligned}\varepsilon_y &= -\frac{\nu\sigma_x}{E} + \frac{\sigma_y}{E} - \frac{\nu\sigma_z}{E} \\ &= -1.067 \times 10^{-3} \text{ in./in.}\end{aligned}$$

$$\begin{aligned}\varepsilon_z &= -\frac{\nu\sigma_x}{E} - \frac{\nu\sigma_y}{E} + \frac{\sigma_z}{E} \\ &= +1.600 \times 10^{-3} \text{ in./in.}\end{aligned}$$

$$\delta_{B/A} = \varepsilon_x d = (+0.533 \times 10^{-3} \text{ in./in.})(9 \text{ in.})$$

$$\boxed{\delta_{B/A} = +4.8 \times 10^{-3} \text{ in.}}$$

$$\delta_{C/D} = \varepsilon_z d = (+1.600 \times 10^{-3} \text{ in./in.})(9 \text{ in.})$$

$$\boxed{\delta_{C/D} = +14.4 \times 10^{-3} \text{ in.}}$$

$$\delta_t = \varepsilon_y t = (-1.067 \times 10^{-3} \text{ in./in.})(0.75 \text{ in.})$$

$$\boxed{\delta_t = -0.800 \times 10^{-3} \text{ in.}}$$

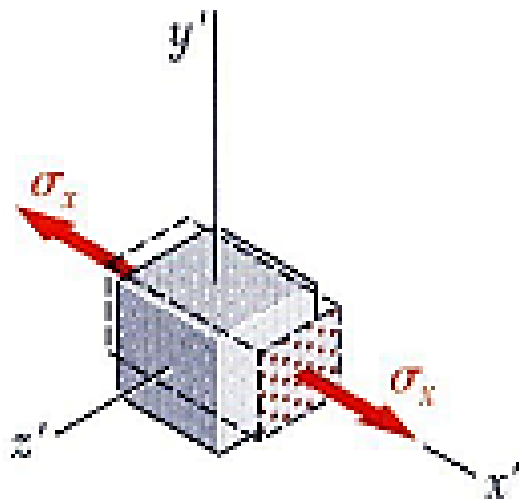
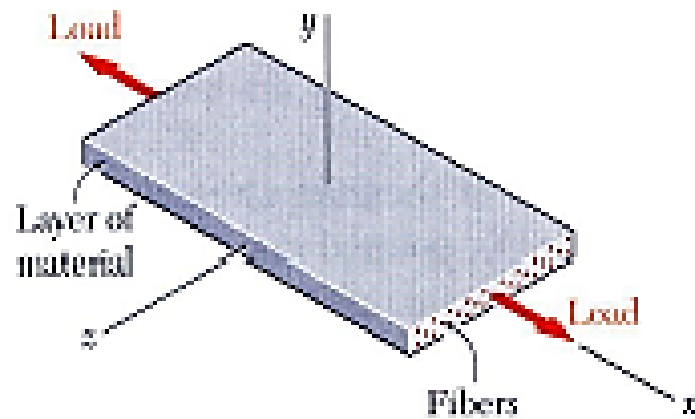
- Find the change in volume

$$e = \varepsilon_x + \varepsilon_y + \varepsilon_z = 1.067 \times 10^{-3} \text{ in}^3/\text{in}^3$$

$$\Delta V = eV = 1.067 \times 10^{-3} (15 \times 15 \times 0.75) \text{ in}^3$$

$$\boxed{\Delta V = +0.187 \text{ in}^3}$$

Composite Materials



- *Fiber-reinforced composite materials* are formed from *lamina* of fibers of graphite, glass, or polymers embedded in a resin matrix.
- Normal stresses and strains are related by Hooke's Law but with directionally dependent moduli of elasticity,

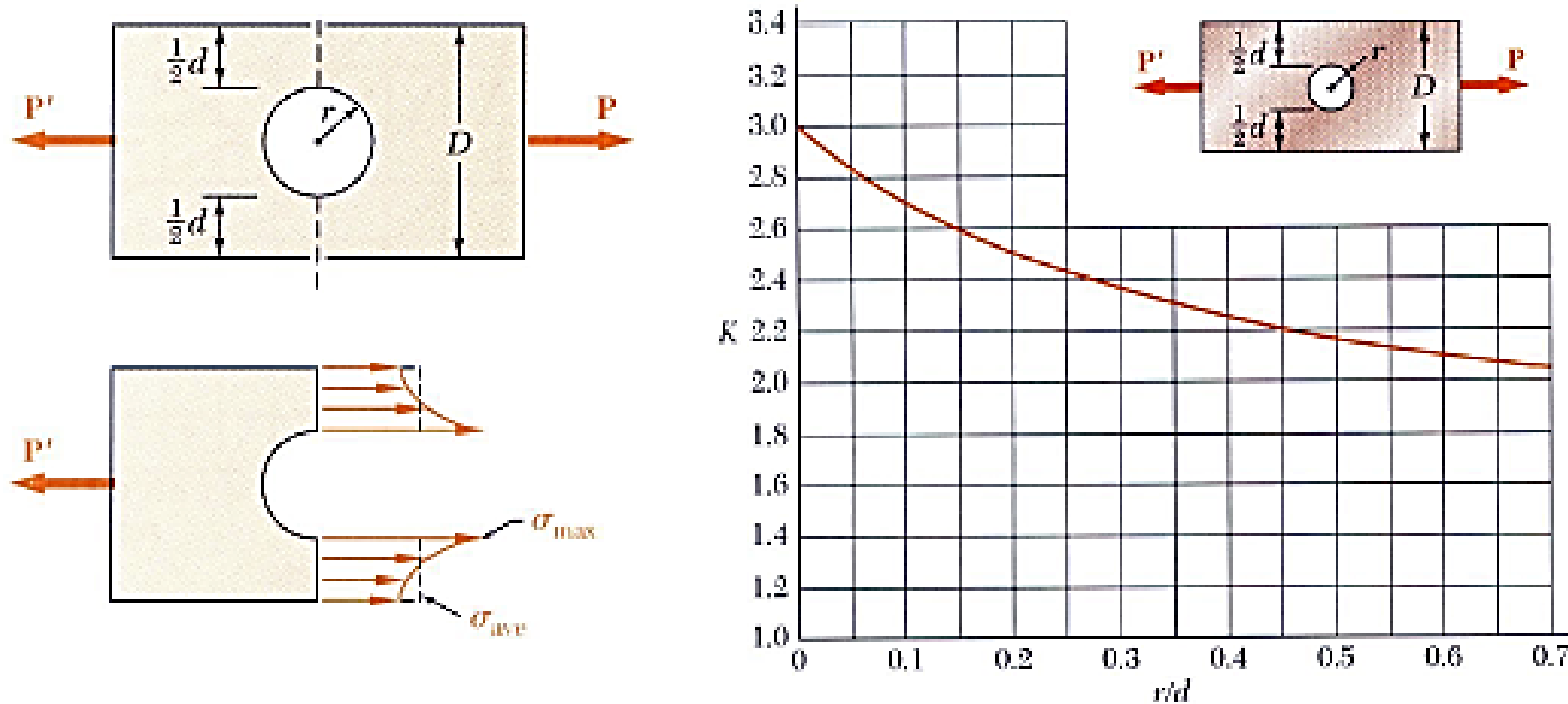
$$E_x = \frac{\sigma_x}{\epsilon_x} \quad E_y = \frac{\sigma_y}{\epsilon_y} \quad E_z = \frac{\sigma_z}{\epsilon_z}$$

- Transverse contractions are related by directionally dependent values of Poisson's ratio, e.g.,

$$\nu_{xy} = -\frac{\epsilon_y}{\epsilon_x} \quad \nu_{xz} = -\frac{\epsilon_z}{\epsilon_x}$$

- Materials with directionally dependent mechanical properties are *anisotropic*.

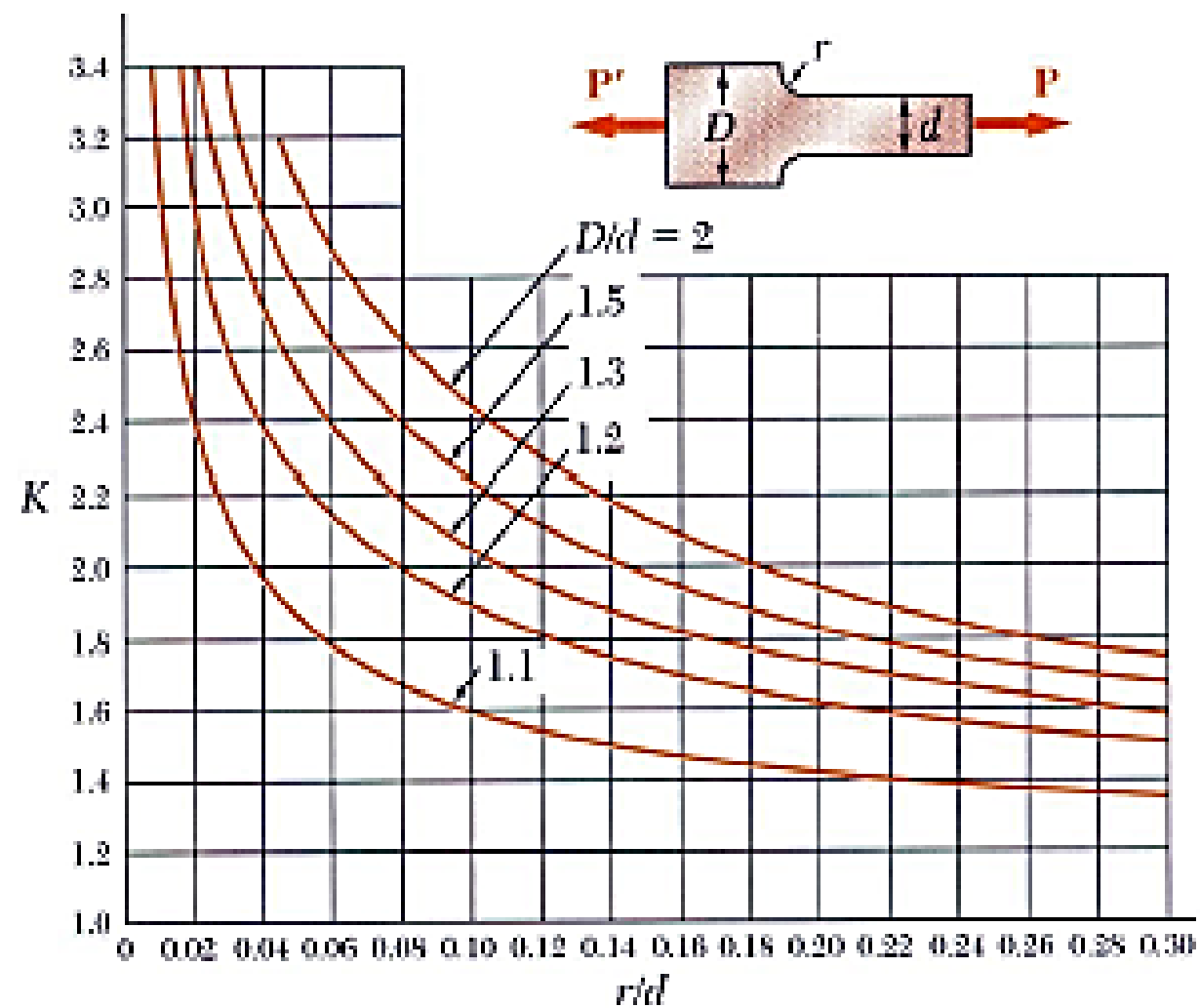
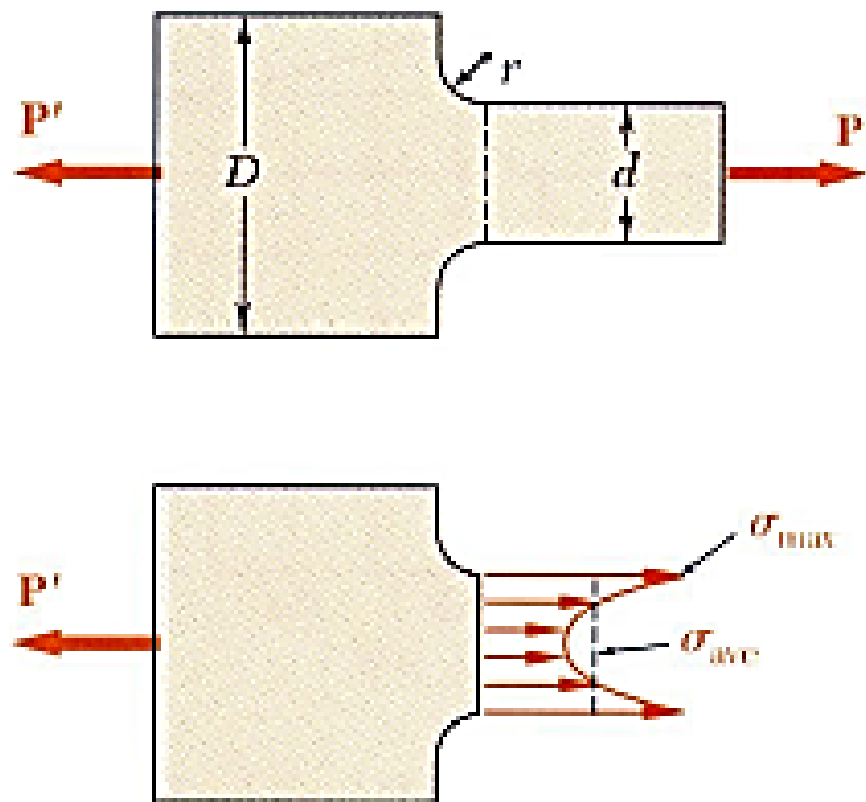
Stress Concentration: Hole



(a) Flat bars with holes

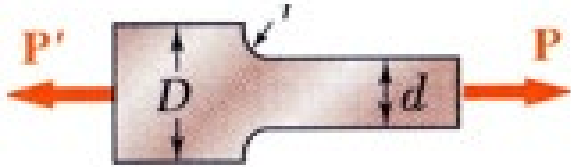
Discontinuities of cross section may result in high localized or *concentrated* stresses.

$$K = \frac{\sigma_{\max}}{\sigma_{\text{ave}}}$$



(b) Flat bars with fillets

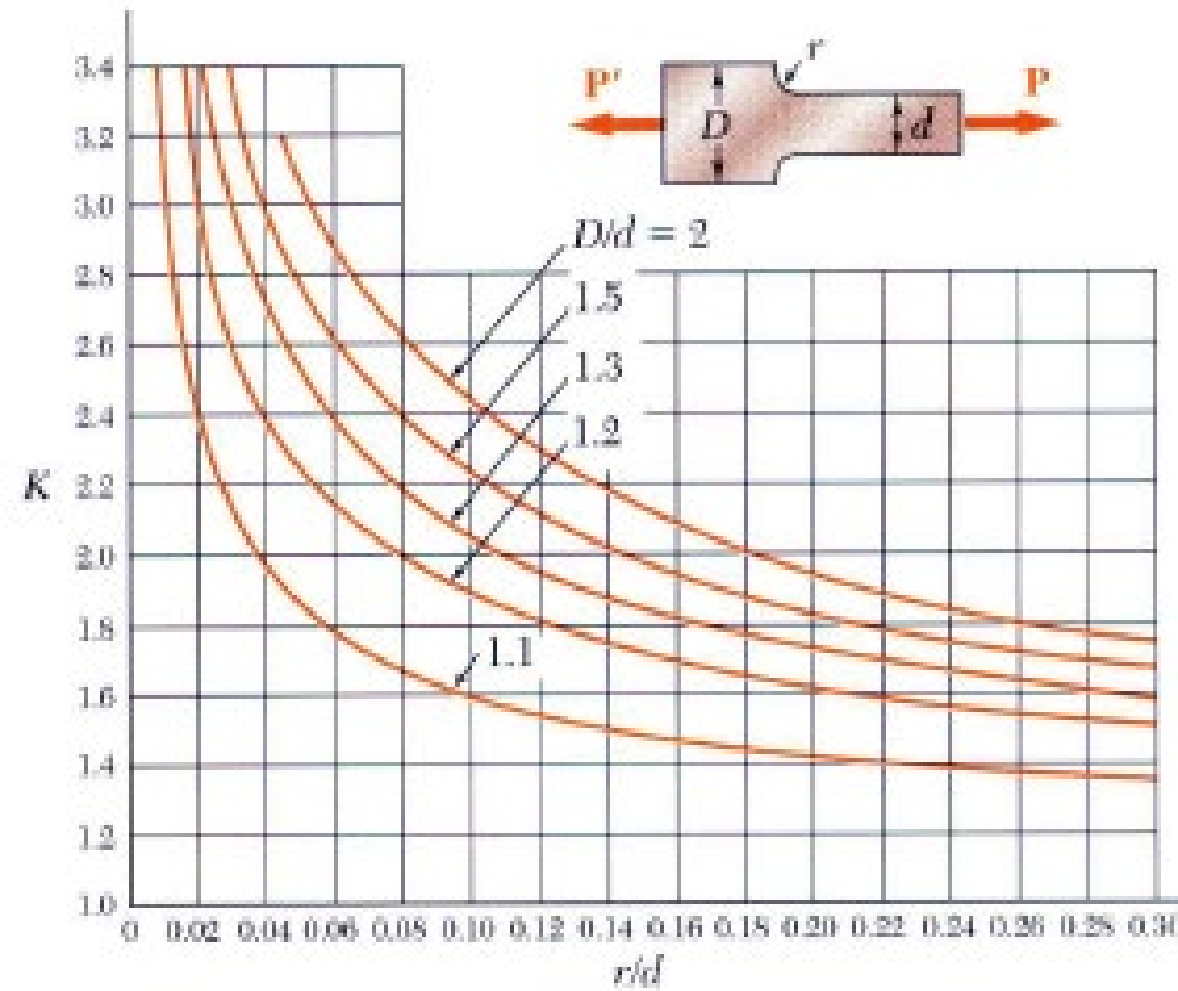
Example



Determine the largest axial load P that can be safely supported by a flat steel bar consisting of two portions, both 10 mm thick, and respectively 40 and 60 mm wide, connected by fillets of radius $r = 8$ mm. Assume an allowable normal stress of 165 MPa.

SOLUTION:

- Determine the geometric ratios and find the stress concentration factor from Fig. 2.64b.
- Find the allowable average normal stress using the material allowable normal stress and the stress concentration factor.
- Apply the definition of normal stress to find the allowable load.



(b) Flat bars with fillets

- Determine the geometric ratios and find the stress concentration factor from Fig. 2.64b.

$$\frac{D}{d} = \frac{60 \text{ mm}}{40 \text{ mm}} = 1.50 \quad \frac{r}{d} = \frac{8 \text{ mm}}{40 \text{ mm}} = 0.20$$

$$K = 1.82$$

- Find the allowable average normal stress using the material allowable normal stress and the stress concentration factor.

$$\sigma_{\text{ave}} = \frac{\sigma_{\text{max}}}{K} = \frac{165 \text{ MPa}}{1.82} = 90.7 \text{ MPa}$$

- Apply the definition of normal stress to find the allowable load.

$$P = A \sigma_{\text{ave}} = (40 \text{ mm})(10 \text{ mm})(90.7 \text{ MPa})$$

$$= 36.3 \times 10^3 \text{ N}$$

$$P = 36.3 \text{ kN}$$

Residual Stresses

- When a single structural element is loaded uniformly beyond its yield stress and then unloaded, it is permanently deformed but all stresses disappear. This is not the general result.
- *Residual stresses* will remain in a structure after loading and unloading if
 - only part of the structure undergoes plastic deformation
 - different parts of the structure undergo different plastic deformations
- Residual stresses also result from the uneven heating or cooling of structures or structural elements



Thank you

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